

Effective-temperature multipoles for relativistic transport.

I. Tangency criterion and entropy-projection inequality

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Relativistic transport in neutrino-thick environments and radiating flows requires solving a high-dimensional Boltzmann equation whose angular structure is expensive to resolve. We parameterise angular nonequilibrium by a direction-dependent effective temperature $\Theta(e)$, expanded in projected symmetric trace-free multipoles on the momentum sphere: along each ray the energy spectrum retains a fixed equilibrium shape (Maxwell–Boltzmann, Fermi–Dirac, or Bose–Einstein), while the angular deformation is encoded in a finite set of Teff multipole coefficients. A low-order truncation of Θ generates, through nonlinear composition, an infinite multipole tail in the standard moments and maps to a closure *region* in moment space rather than a single curve. To detect when this ray-wise-equilibrium ansatz ceases to be collision-side tangent, we project the entropy-normalised collision field onto a generalised Laguerre basis in energy. The resulting tangency criterion—an instantiation of Gorbán and Karlin’s defect of invariance for per-direction exponential families—vanishes if and only if the collision forcing is tangent to the enlarged per-ray two-parameter family $\mathcal{M}_{\text{ray}}$ (not the full Teff manifold; the angular-consistency condition on the constant mode $n = 0$ is a separate obstruction, Remark VB). An entropy-projection inequality decomposes the reduction-map entropy cost (a single-instant, reduction-map statement, not a dynamical H-theorem) into a shape-mismatch term bounded by the $n \geq 2$ energy modes of the state residual and an amplitude-mismatch term; the proof applies the standard Pythagorean identity for Bregman divergences to the per-ray exponential family. The collision diagnostic monitors the same Laguerre sector on a different object (collision field vs. state deviation; Theorem 11); it predicts the rate at which the entropy gap opens, not the current gap itself. The tangency criterion is a per-ray, instantaneous, collision-side energy-mode test; it is necessary but not by itself sufficient for full Teff-manifold tangency, and it does not guarantee global kinetic adequacy. All formal results assume $\mu \leq 0$ (equivalently, $y > 0$), which covers MB, BE, and non-degenerate FD; the degenerate extension $\mu > 0$ is discussed but not proved. The diagnostic is verified on controlled collision operators: machine-precision tangency for all three statistics under multipole relaxation, quantified non-tangency for quantum statistics under BGK isoennerisation with a Bose–Einstein/Fermi–Dirac diagnostic ratio set by a computable spectral integral (≈ 0.66 at leading order), and Thomson scattering tangent at linear order with $O(A^2)$ correction. Paper II develops the direction-dependent chemical-potential extension; Paper III establishes the entropy geometry of the extended manifold.

I. INTRODUCTION

Relativistic transport in neutrino-thick environments—core-collapse supernovae, proto-neutron-star cooling, neutron-star mergers [1–3]—and in photon radiation hydrodynamics requires evolving a phase-space distribution $f(x^a, p^a)$ that depends on space, energy, and angle. The angular structure is typically the dominant computational obstacle [4, 5]. Two-moment (M_1) closures, which evolve only the energy density and flux, are the de facto standard in multidimensional simulations; they require a single closure relation but suffer from well-documented structural failures in multi-source and beam-crossing configurations [6, 7]. Filtered spherical-harmonic (FP_N) methods increase angular resolution systematically but at steep computational cost and with filter-dependent artefacts in the optically thin limit [8, 9]. The question that motivates this

work is not whether to reduce the angular degrees of freedom, but which reduced variables compress angular structure while providing a clear contract for when the compression fails.

We parameterise angular nonequilibrium by a direction-dependent effective temperature $\Theta(e)$ on the momentum sphere. Along each ray (fixed direction e in the local rest space), the energy spectrum retains a prescribed equilibrium shape—Maxwell–Boltzmann, Fermi–Dirac, or Bose–Einstein—with a temperature that varies continuously across the sphere:

$$f_{\Theta}(y, e) = \Phi_{\xi}\left(\frac{y}{\Theta(e)}\right), \quad y := \frac{E - \mu}{k_B T_0}. \quad (1)$$

The field $\Theta(e)$ is expanded in projected symmetric trace-free (PSTF) multipoles on the sphere with the monopole locked to unity ($\langle \Theta \rangle_{\Omega} = 1$); the dynamical degrees of freedom are the multipole coefficients T_{A_ℓ} for $\ell \geq 1$. The notion of a direction-dependent effective temperature has a long history in nonequilibrium statistical physics [10, 11], where it encodes the departure of a slow subsystem from global equilibrium; in the present context $\Theta(e)$ serves the analogous role of encoding the departure of the angular

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distribution from isotropy.

Existing angular reduction strategies fall into three families, each with a characteristic limitation. Maximum-entropy moment closures [12–14] select the distribution that maximises entropy at fixed low-order moments; they yield realisable, hyperbolic systems but require solving a constrained optimisation problem at each spacetime point, and the conditioning degenerates near the realisability boundary—a pathology analysed by Junk [15] and further characterised by Hauck, Levermore, and Tits [16]. Partial-moment closures [17] assign independent entropy closures to angular patches of S^2 , gaining per-direction resolution but discretising the angular domain and losing the continuous PSTF structure. At the abstract-kinetic level, the defect-of-invariance framework of Gorban and Karlin [18–20] provides the conceptual tool for detecting when a reduced manifold ceases to be dynamically invariant; however, it has been developed for gas kinetics and polymer dynamics, not for radiation transport with its per-direction spectral families. Recent work by Abdelmalik, Cai, and Pichard [21] applies φ -divergence closures to the radiative transfer equation using global angular moment constraints. To the best of our knowledge, what has not been combined in a single framework is a per-direction exponential-family parameterisation with an intrinsic diagnostic that monitors, at each collision evaluation, whether the collision forcing remains tangent to the reduced manifold (Proposition 7 delimits the scope of this diagnostic).

The present framework supplies that combination. We restrict the distribution to the ray-wise equilibrium family (1)—a one-parameter exponential family per direction—and construct two diagnostic tools. The first is a tangency criterion (Theorem 6): we project the entropy-normalised collision field $G_\xi = C/[f(1+\xi f)]$ onto a generalised Laguerre basis in energy and read off the higher-mode content ($n \geq 2$). These modes vanish if and only if the collision forcing is tangent to the enlarged per-ray two-parameter family $\mathcal{M}_{\text{ray}}$; reaching the global Teff manifold requires, in addition, the angular-consistency condition on the constant mode $c_0(e)$. Their norm defines the collision-side diagnostic scalar. The criterion is an instantiation of Gorban and Karlin’s defect of invariance for per-direction exponential families (Sec. V B), with three operational modifications detailed in Sec. V. The second tool is an entropy-projection inequality (Theorem 9): the proof applies the standard Pythagorean identity for Bregman divergences on the per-ray exponential family. The resulting inequality is a single-instant reduction-map statement, not a dynamical H -theorem; it is controlled by the $n \geq 2$ energy modes of the *state residual* $\mathcal{D}_{\geq 2}^{\text{res}}$. The collision diagnostic $\mathcal{D}_{\geq 2}$, computed from the same Laguerre sector on a different object (the collision field rather than the state residual), predicts the leading-order rate at which the shape-mismatch component of the reduction-map entropy gap opens; it is not a measurement of the current

gap itself (Theorem 10, Theorem 11); the total entropy gap includes an additional per-ray amplitude-mismatch term. The entropy bound itself is a reduction-map statement, not a time-evolution theorem, and the angular-consistency condition on the $n = 0$ sector is separate (Remark V B, Proposition 7(d,e)).

Three structural features distinguish the present work from existing reduced descriptions. First, the per-direction Planckian closure $\Phi(y/\Theta)$ is explicit—it requires no optimisation and avoids the boundary conditioning pathology of global entropy closures [15, 16, 22]. A low-order truncation of Θ generates, through nonlinear composition, an infinite multipole tail in the standard moments and maps to a closure *region* in moment space rather than a single curve. Second, the tangency diagnostic triggers a qualitative model switch—activation of energy-shape modes or a multigroup transition—rather than iterative manifold correction, and its evaluation reduces to weighted energy integrals already computed for the hierarchy update (the cost advantage over the Gorban–Karlin Newton correction lies primarily in the response mechanism—model switch versus iterative refinement—rather than in the diagnostic evaluation itself, which is comparably inexpensive for low-dimensional manifolds in both frameworks). We are not aware of a prior radiation moment method that provides an in-situ, collision-side tangency diagnostic for a prescribed per-direction exponential-family manifold, together with an explicit entropy/projection bound controlled by the same off-manifold modes (a dedicated literature search may reveal precedents we have not identified). Existing realisability-preserving schemes [23] enforce admissibility constraints ($|F| \leq cE$) but do not quantify collision-induced departure from a chosen distribution manifold in energy-mode coordinates. Third, the dually flat structure of the per-ray exponential family [24] unifies the diagnostic and the entropy gap through the Fisher metric: the same entropy Hessian $1/[f(1+\xi f)]$ that straightens the tangent space for the diagnostic also weights the reduction-map information cost [25–27] (Corollary 13).

The paper makes five contributions. **C1** (Representation, Sec. III). We introduce Teff multipoles as a nonlinear angular parameterisation of relativistic distributions, characterise the induced high- ℓ tail, and derive the exact covariant frame-transformation law (for $\mu = 0$; Proposition 4) with its boost-induced multipole cascade. **C2** (Tangency diagnostic, Sec. V). We formulate the per-ray tangency criterion (Theorem 6), prove Thomson scattering tangent at linear order with second-order correction scaling as the angular variance of Θ , and design a scalar diagnostic whose nonvanishing signals that the collision forcing is not absorbed by updates of the per-ray temperature and per-ray normalisation (Proposition 7). **C3** (Entropy structure, Sec. VI). We identify the per-ray family as a one-parameter exponential family, establish an entropy-projection inequality (Theorem 9) bounding the reduction-map entropy cost by the state-residual diagnostic $\mathcal{D}_{\geq 2}^{\text{res}}$, with the collision diagnostic $\mathcal{D}_{\geq 2}$

predicting the opening rate (Corollary 13 delimits the scope), and derive sharp realisability certificates for the mixed dipole-plus-quadrupole truncation. **C4** (Structural verification, Sec. VII). We verify the representation, diagnostic, and realisability results on controlled collision operators (Table II): machine-precision tangency for all three statistics under multipole relaxation, quantified non-tangency for Fermi–Dirac and Bose–Einstein under BGK isoenergisation with a diagnostic ratio set by a statistics-dependent spectral integral (≈ 0.66 at leading order), and Thomson tangency at linear order. The tangent-direction verification is now symmetric across statistics; the non-tangent benchmarks use controlled operators (not yet tested on microphysical kernels).

This work is Teff-only: we exclude spectral distortions by construction, and we use a tangency diagnostic to trigger upgrades (an added energy-shape mode or a multigroup switch) when Teff-only ceases to be self-consistent. All benchmarks use controlled collision operators in local-frame or axisymmetric settings to isolate the representation and diagnostic mechanisms. Quantitative benchmarks against reference transport solutions—including a variable-Eddington-factor comparison, extensions to massive and degenerate gases, and the symmetric-hyperbolic structure of the projected hierarchy—are deferred to future work.

Standalone contract. The present paper proves all formal results (Theorems 3–10, Propositions 5–15) and verifies them on controlled operators (Table II). The formal content is self-contained: no result depends on the subsequent papers in the series. Paper II extends the manifold to include a direction-dependent chemical potential; Paper III develops the entropy geometry of the extended manifold.

Sec. II fixes notation, the 1+3 split, and the momentum-sphere variables. Sec. III introduces the Teff manifold, the induced tail, and the closure-region geometry. Sec. IV derives the projected hierarchy and characteristic speeds. Sec. V formulates the tangency diagnostic, the Gorbán–Karlin connection, and the Thomson tangency analysis. Sec. VI establishes the exponential-family structure, the entropy-projection inequality, and the realisability certificates. Sec. VII presents the structural verifications. Sec. VIII discusses the validity regime, broader applicability, and limitations. Appendices A and B collect proofs and projection identities.

II. COVARIANT SETUP AND NOTATION

We work on a general four-dimensional spacetime $(\mathcal{M}, g_{ab})$ with metric signature $(-, +, +, +)$ and Levi–Civita connection ∇_a . A timelike unit observer field u^a satisfies $u^a u_a = -1$ and defines the spatial projector

$$h_{ab} := g_{ab} + u_a u_b, \quad (2)$$

which obeys $h_{ab} u^b = 0$ and $h^a_a = 3$. Indices are raised and lowered with g_{ab} . From now on, we use the standard

dot product $u \cdot p := u_a p^a$ and denote convective time derivatives by an overdot.

A. 1+3 split and PSTF projection

We adopt the covariant 1+3 threading formalism [28–30]. For any scalar X the convective and projected derivatives are

$$\dot{X} := u^a \nabla_a X, \quad D_a X := h_a^b \nabla_b X. \quad (3)$$

Round brackets $(\dots)$ and square brackets $[\dots]$ denote symmetrization and antisymmetrization, respectively. Angle brackets on free indices always mean “projected, symmetric, trace-free” (PSTF) with respect to u^a . For a covector A_a and a rank-2 tensor A_{ab} we define

$$A_{\langle a} := h_a^b A_b, \quad (4)$$

$$A_{\langle ab \rangle} := h_{(a}^c h_{b)}^d A_{cd} - \frac{1}{3} h_{ab} h^{cd} A_{cd}. \quad (5)$$

For higher ranks we write a multi-index $A_\ell \equiv a_1 \dots a_\ell$ and denote the PSTF part by $X_{\langle A_\ell \rangle}$.

When needed, we encode the kinematics of the observer congruence by the standard decomposition

$$\nabla_b u_a = \frac{1}{3} \theta h_{ab} + \sigma_{ab} + \omega_{ab} - u_b a_a, \quad (6)$$

where $\theta := D_a u^a$ is the expansion, $\sigma_{ab} := D_{\langle a} u_{b \rangle}$ the shear, $\omega_{ab} := D_{[a} u_{b]}$ the vorticity, and $a^a := \dot{u}^a$ the acceleration. The expansion, shear, and vorticity play no role until the projected hierarchy is discussed in Sec. IV; we record them here to keep the kinematic decomposition in one place.

B. Momentum variables and invariant measure

Our goal in this subsection is to fix the energy–angle split of the four-momentum and the on-shell integration measure that all subsequent integrals refer to.

Let p^a denote the on-shell particle four-momentum with rest mass m ,

$$p^a p_a = -m^2. \quad (7)$$

Relative to u^a we decompose p^a into an energy and a spatial direction,

$$p^a = E u^a + \lambda e^a = E (u^a + v e^a), \quad (8)$$

where the observer-frame energy is

$$E := -u_a p^a \quad (> 0), \quad (9)$$

the spatial direction satisfies $u_a e^a = 0$ and $e_a e^a = 1$, and

$$\lambda := \sqrt{E^2 - m^2}, \quad v := \lambda/E. \quad (10)$$

For massless particles, $\lambda = E$ and $v = 1$; in this case the physical energy domain is $E \in (0, \infty)$, while for massive particles $E \geq m > 0$. The sign convention in Eq. (9) (positive E for a future-directed p^a) follows directly from $u^a u_a = -1$ and p^a future-directed; it is consistent with the momentum split (8) via $u_a p^a = E u_a u^a + \lambda u_a e^a = -E$.

The Lorentz-invariant on-shell measure in these local $1 + 3$ variables is

$$dP \equiv \frac{d^3 p}{p^0} = \frac{\lambda^2}{E} d\lambda d\Omega, \quad (11)$$

where $d\Omega$ is the solid-angle element on the unit sphere of directions e^a in the local rest space of u^a . When forming invariant integrals for currents and stress-energy, we include the usual degeneracy and quantum prefactor by replacing $dP \mapsto \frac{g}{(2\pi\hbar)^3} dP$; the degeneracy factor g counts internal degrees of freedom (spin, helicity) and is absorbed into the overall normalization of the macroscopic fields.

C. Distribution, statistics, and collision term

We describe the medium by a one-particle distribution $f(x, p)$ interpreted as an occupation number [31, 32]. To accommodate Bose enhancement and Pauli blocking in a unified notation we introduce the statistics sign

$$\xi = \begin{cases} 0 & \text{classical (Boltzmann limit),} \\ +1 & \text{bosons (Bose-Einstein),} \\ -1 & \text{fermions (Fermi-Dirac),} \end{cases} \quad (12)$$

so that quantum factors are written as $(1 + \xi f)$. The admissible domain is $f \geq 0$ for bosons, $0 \leq f \leq 1$ for fermions, and $f \ll 1$ for the classical limit.

The covariant kinetic equation is

$$\mathcal{L}[f](x, p) = C[f](x, p), \quad (13)$$

where $\mathcal{L}$ is the Liouville operator on the mass shell (including geometry and possible external forces) and C is a local, Markovian collision operator. The precise gain-loss structure (e.g., Uehling-Uhlenbeck for quantum $2 \leftrightarrow 2$ scattering) is introduced only when needed in Secs. IV and V; all kinematic terms remain on the left-hand side of Eq. (13). This deliberate delay keeps the present section collision-model-agnostic: the covariant setup, the PSTF basis, and the multipole normalization hold regardless of the form of C .

D. PSTF multipoles on the momentum sphere

Following the conventions of Thorne [33, 34], for any scalar function of direction on the momentum sphere—and in particular for f at fixed energy—we use a PSTF

multipole expansion,

$$f(x, E, e) = \sum_{\ell=0}^{\infty} F_{A_\ell}(x, E) e^{A_\ell}, \quad (14)$$

$$e^{A_\ell} := e^{\langle a_1 \dots e^{a_\ell} \rangle},$$

where $F_{A_\ell} = F_{\langle A_\ell \rangle}$ are spatial PSTF tensors. The coefficients are recovered by angular projection,

$$F_{A_\ell}(x, E) = \frac{1}{\Delta_\ell} \int d\Omega f(x, E, e) e_{\langle A_\ell \rangle},$$

$$\Delta_\ell := \frac{4\pi 2^\ell (\ell!)^2}{(2\ell + 1)!}. \quad (15)$$

The normalization constant Δ_ℓ is fixed by the orthogonality relation for PSTF harmonics (the lowest values are $\Delta_0 = 4\pi$, $\Delta_1 = 4\pi/3$, $\Delta_2 = 8\pi/15$). These definitions lock the normalization of all multipoles used below, including the effective-temperature multipoles introduced in Sec. III and the projected collision multipoles in Sec. IV.

III. EFFECTIVE-TEMPERATURE MULTIPOLES AS A NONLINEAR ANGULAR REPRESENTATION

A. Per-ray equilibrium manifold and Teff multipoles

We now define the distribution-level representation that underlies the effective-temperature (Teff) hierarchy. Throughout we assume the covariant $1 + 3$ split and PSTF angular basis (4)–(5) fixed in Sec. II, so that a future-directed four-momentum decomposes as $p^a = E u^a + \lambda e^a$ (10) with $e^a u_a = 0$ and $e^a e_a = 1$. Our goal in this section is not to derive any evolution equation, but to make explicit what is being parameterized and what is not.

To this end, we restrict the occupation-number distribution to a ray-wise equilibrium family,

$$f_\Theta(x, p) \equiv f_\Theta(y, e) = \Phi\left(\frac{y}{\Theta(x, e)}\right), \quad y := \frac{E - \mu(x)}{k_B T_0(x)}. \quad (16)$$

Here $T_0(x)$ is a reference temperature scale, $\mu(x)$ is a chemical potential ($\mu = 0$ for MB and BE; for FD, $\mu \in \mathbb{R}$ in general, but the present diagnostic construction restricts to $\mu \leq 0$ as a sufficient convention ensuring $y > 0$ and the Laguerre basis on $(0, \infty)$; for massive species, weaker conditions ($\mu < m$) can also ensure $y > 0$, but we do not pursue this here (see Sec. V), and Φ is the equilibrium spectral shape appropriate to the chosen statistics: Maxwell-Boltzmann (MB), Bose-Einstein (BE), or Fermi-Dirac (FD). The angular dependence enters only through the positive scalar field $\Theta(x, e)$ on the unit momentum sphere, which rescales the argument of Φ along each direction. A quadrupole-level version of the

direction-dependent temperature ansatz was introduced in a cosmological setting by [35]; the present work extends this to general truncation order, equips it with a tangency diagnostic, and develops the entropy structure for general UU-consistent transport applications.

By construction, Eq. (16) is Teff-only: it cannot represent genuine spectral distortions, meaning any independent change of the functional shape in y at fixed direction e beyond the rescaling $y \mapsto y/\Theta$. We treat this scope restriction as a design commitment, not a limitation to be apologized for: it is what makes the tangency diagnostic (Sec. V) a yes/no diagnostic rather than a graded approximation. When collisions generate energy-shape content that violates this restriction, we do not “patch” the ansatz; instead, we diagnose that failure and switch to an upgraded model (Sec. V).

The split into an isotropic scale $T_0(x)$ and an angular deformation $\Theta(x, e)$ has a local scaling redundancy: $T_0 \mapsto \lambda T_0$ and $\Theta \mapsto \Theta/\lambda$ leave y/Θ (hence f_Θ) invariant. We fix this redundancy by locking the monopole of Θ ,

$$\langle \Theta \rangle_\Omega = 1, \quad T_{\text{eff}}(x, e) = T_0(x) \Theta(x, e),$$

$$T_0(x) = \frac{1}{4\pi} \int d\Omega T_{\text{eff}}(x, e), \quad (17)$$

where $\langle \cdot \rangle_\Omega$ denotes the angular mean in the comoving rest space of u^a . The gauge condition $\langle \Theta \rangle_\Omega = 1$ holds to machine precision ($\sim 10^{-15}$) under Gauss–Legendre quadrature for all realizable multipole combinations tested (see Appendix B for numerical details). The dynamical degrees of freedom are then the PSTF multipoles of Θ (with $\ell \geq 1$),

$$\Theta(x, e) = 1 + \theta(x, e),$$

$$\theta(x, e) = \sum_{\ell=1}^{\infty} T_{A_\ell}(x) e^{A_\ell}, \quad e^{A_\ell} := e^{a_1 \dots e^{a_\ell}}. \quad (18)$$

In computations we truncate (18) at some finite L ,

$$\Theta^{(L)}(x, e) = 1 + \sum_{\ell=1}^L T_{A_\ell}(x) e^{A_\ell}, \quad (19)$$

and evolve only the fields $T_{A_\ell}(x)$, $\ell \leq L$. At fixed L , any desired moment is a derived quantity obtained by integrating $f_{\Theta^{(L)}}$ against the invariant measure; symbolically this defines a nonlinear “moment map”

$$\{T_{A_\ell}\}_{\ell=1}^L \longmapsto \text{moments of } f_{\Theta^{(L)}}. \quad (20)$$

It is worth to note that this is not a moment closure in the usual sense: we do not prescribe an algebraic relation among a small subset of low-order moments. We prescribe instead a finite-dimensional *manifold* in distribution space, and then compute moments as functionals on that manifold. The distinction is structural. A closure answers the question “given $(\mathcal{E}, \mathcal{F}^a)$, what is P^{ab} ?” by an algebraic curve. The Teff manifold answers a different question: “given $\{T_{A_\ell}\}_{\ell \leq L}$, what is f ?” and from

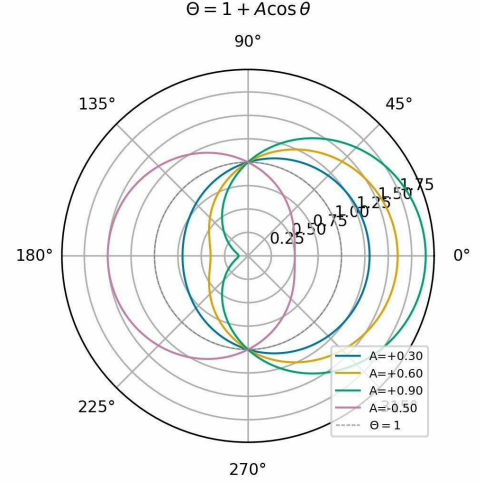


FIG. 1. Polar plot of the dipole Teff deformation $\Theta(\theta) = 1 + A \cos \theta$ for amplitudes $A = +0.30$ (blue), $+0.60$ (orange), $+0.90$ (teal), and -0.50 (pink), together with the isotropic reference $\Theta = 1$ (dotted grey circle). The radial coordinate represents the effective temperature ratio $\Theta = T_{\text{eff}}/T_0$; the forward direction ($\theta = 0$) points right. For positive A , rays propagating in the forward hemisphere have enhanced T_{eff} , mimicking a beamed radiation field. At $A = +0.90$, the backward hemisphere nearly reaches the realizability limit $\Theta \rightarrow 0$. For $|A| < 1$, realizability ($\Theta > 0$ for all directions) is automatically satisfied.

that f , *all* moments follow. In Sec. V we exploit this viewpoint: the kinetic dynamics itself tells us when the trajectory is leaving the Teff-only manifold and which type of enrichment is required.

Figures 1 and 2 illustrate the geometric content of these deformations. At dipole level, $\Theta(e) = 1 + T_a e^a$ describes a cardioid-like deformation of the effective-temperature surface on the momentum sphere (Fig. 1): forward-peaked rays carry a higher spectral temperature while backward rays are cooled, mimicking a beamed radiation field within the ray-wise equilibrium framework. Adding a quadrupole, $\Theta(e) = 1 + T_a e^a + T_{ab} e^a e^b$, permits prolate or oblate deformations whose axisymmetric cross-section and full Mollweide projection are shown in Fig. 2. Both panels confirm that the realizability constraint $\Theta(e) > 0$ is satisfied everywhere on the sphere for the displayed parameter values; the formal characterization of the realizable region is given in Sec. VID.

B. Why a low-order Θ induces an infinite multipole tail in f

The Teff truncation in Eq. (19) should not be confused with a truncation of the standard angular multipoles of the distribution itself. For any scalar $X(x, E, e)$ on the momentum sphere we define the usual PSTF multipoles

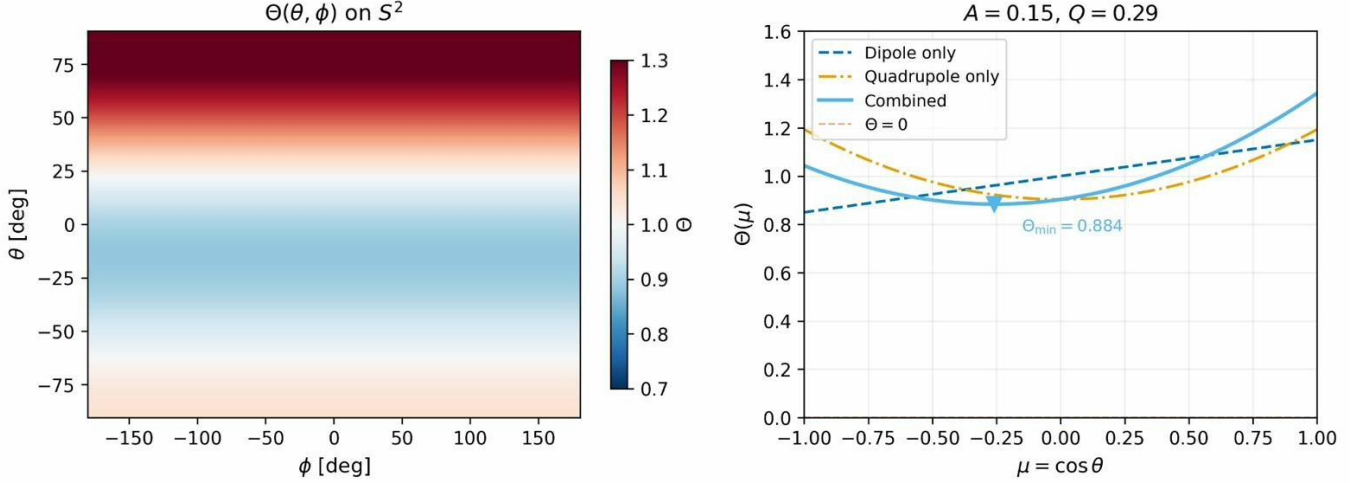


FIG. 2. Quadrupole Teff deformation with $A = 0.15$, $Q = 0.29$. (Left) Mollweide projection of $\Theta(\theta, \phi)$ on the unit momentum sphere. Because $Q > 0$, the effective temperature is enhanced along the polar axis and suppressed in the equatorial plane; the dipole $A > 0$ breaks forward-backward symmetry. The colour scale ranges from $\Theta \approx 0.7$ (dark blue, near $\theta = \pi$) to $\Theta \approx 1.3$ (dark red, near $\theta = 0$). (Right) Axisymmetric cross-section $\Theta(\mu)$ with $\mu = \cos \theta$, showing the individual dipole-only (dashed blue), quadrupole-only (dot-dashed orange), and combined (solid teal) contributions. The minimum $\Theta_{\min} = 0.884$ (teal triangle) lies well above the realizability boundary $\Theta = 0$ (dashed pink).

by

$$X(x, E, e) = \sum_{\ell=0}^{\infty} X_{A_\ell}(x, E) e^{A_\ell},$$

$$X_{A_\ell}(x, E) = \frac{1}{\Delta_\ell} \int d\Omega X(x, E, e) e_{\langle A_\ell \rangle}, \quad (21)$$

with Δ_ℓ the standard normalization factor fixed in Sec. II. Applying Eq. (21) to $X = f_\Theta$ defines the standard multipoles $F_{A_\ell}(x, E)$ of the distribution at fixed energy.

Now comes the structural point: even if Θ is truncated at $\ell \leq L$, the composed function $e \mapsto \Phi(y/\Theta(e))$ is generically not a spherical polynomial of degree L . Therefore the induced multipoles $F_{A_\ell}(x, E)$ exhibit an infinite high- ℓ tail. To see the mechanism without a long computation, write $\Theta = 1 + \theta$ and formally expand

$$\frac{1}{\Theta} = \frac{1}{1 + \theta} = \sum_{n=0}^{\infty} (-1)^n \theta^n, \quad (|\theta| < 1). \quad (22)$$

For Maxwell-Boltzmann statistics $\Phi(z) = e^{-z}$, inserting (22) gives $e^{-y/\Theta} = e^{-y(1-\theta+\theta^2-\dots)}$. If θ contains multipoles only up to rank L , then θ^n contains angular harmonics up to rank nL . Hence each order in the expansion mixes multipole ranks, and for finite anisotropy the series generates nonzero content at arbitrarily large ℓ . For quantum statistics one may use the standard exponential series $\Phi_{\text{BE/FD}}(z) = \sum_{k \geq 1} \sigma_k e^{-kz}$ (with $\sigma_k = 1$ for BE and $\sigma_k = (-1)^{k+1}$ for FD) to reduce the discussion to Boltzmann-like terms; the conclusion—a generically induced high- ℓ tail—is unchanged. (The geometric series (22) converges only for $|\theta| < 1$, whereas the realizability constraint $\Theta > 0$ allows $\theta \in (-1, \infty)$. The

multipole-rank mixing mechanism is not an artifact of the series: the composed function $\Phi(y/\Theta(e))$ is well defined for all $\Theta > 0$, and exact quadrature—used in all benchmarks—captures the induced tail without relying on the series convergence.)

Remark (Generating function and self-consistency of the MB expansion). For MB statistics the Taylor expansion of $e^{-y/\Theta}$ in powers of θ admits a closed generating function:

$$e^{-y/(1+\theta)} = e^{-y} \sum_{n=0}^{\infty} \hat{P}_n(y) \theta^n, \quad (23)$$

$$\sum_{n=0}^{\infty} \hat{P}_n(y) \theta^n = \exp\left(\frac{y\theta}{1+\theta}\right),$$

where $\hat{P}_n(y) = \sum_{m=1}^n (-1)^{n-m} \binom{n-1}{m-1} y^m / m!$ are reduced polynomials of degree n . This provides a direct verification that the formal series converges to the exponential composition. As a self-consistency check, the angle-averaged distribution $\bar{f}^{(2)}$ at order θ^2 can be obtained by two independent routes: (A) Taylor-expand $e^{-y/(1+\theta)}$ to second order and average term by term using $\langle \theta \rangle_\Omega = 0$, $\langle \theta^2 \rangle_\Omega = \Sigma_2$; or (B) write $e^{-y/(1+\theta)} = e^{-y} e^{y\eta}$ with $\eta = \theta/(1+\theta)$ and apply the cumulant expansion with $\kappa_1 = -\Sigma_2 + \mathcal{O}(\theta^3)$, $\kappa_2 = \Sigma_2 + \mathcal{O}(\theta^3)$. Both yield $\bar{f}^{(2)} = e^{-y} [1 + \frac{1}{2} \Sigma_2 y(y-2)]$, confirming the internal consistency of the expansion.

The correction factor $\frac{1}{2} \Sigma_2 y(y-2)$ in $\bar{f}^{(2)}$ changes sign at $y = 2$: it is negative for $0 < y < 2$ (a deficit of soft particles) and positive for $y > 2$ (an excess of hard particles). The angle-averaged spectrum

is therefore spectrally harder than the isotropic equilibrium $\Phi(y)$ —a redistribution structurally analogous to the thermal Sunyaev–Zel’dovich y -distortion, with the anisotropy variance Σ_2 playing the role of the Comptonization parameter. (The analogy is mathematical—both arise from Jensen’s inequality applied to a temperature mixture—not physical: the anisotropy amplitudes in the primary targets of this work ($\theta \sim 0.01$ – 0.9) exceed CMB anisotropies ($\delta T/T \sim 10^{-5}$) by orders of magnitude.) The crossover at $y = 2$ and its sign structure are determined by Jensen’s inequality applied to $g(\Theta) = e^{-y/\Theta}$: since $g''(1) = e^{-y} y(y-2)$, the map $\Theta \mapsto e^{-y/\Theta}$ is convex at $\Theta = 1$ for $y > 2$ and concave for $y < 2$, giving $\langle e^{-y/\Theta} \rangle_\Omega \geq e^{-y}$ in the hard tail and $\langle e^{-y/\Theta} \rangle_\Omega \leq e^{-y}$ in the soft sector. Thus even at the isotropic-average level, the Teff truncation generates nontrivial spectral corrections that are absent from any linear angular truncation.

We emphasize that this is a statement about angular structure only. It does not reintroduce spectral distortions, which remain excluded by construction in Eq. (16). The induced tail says that Teff truncation is a nonlinear angular compression, not a linear truncation of f at fixed energy. That is, the tail is the *mechanism* by which a small set of Teff parameters captures angular complexity that a linear truncation with the same number of degrees of freedom cannot.

C. Closure region, not closure curve

The induced tail has an immediate implication for moment closures. In two-moment language one often characterizes a closure by an Eddington-factor relation $\chi = \chi(\mathbf{f})$, where $\mathbf{f} := |\mathcal{F}|/\mathcal{E}$ is a flux factor and χ is a scalar summary of the pressure tensor (for axisymmetric configurations, $\chi = P_{||}/\mathcal{E}$). This is a curve in $(\mathbf{f}, \chi)$ space because the closure is parameterized by a single scalar $\mathbf{f}$.

In the Teff representation, the pressure tensor and higher moments are nonlinear functionals of the finite parameter set $\{T_{A_\ell}\}_{\ell \leq L}$, so the implied relation between $(\mathbf{f}, \chi)$ is generically higher-dimensional. To be explicit, define the Teff-induced admissible set

$$\mathcal{C}_{\text{Teff}}^{(L)} := \left\{ (\mathbf{f}(T_{A_\ell}), \chi(T_{A_\ell})) \mid \Theta^{(L)}(e) > 0, T_{A_\ell} \text{ truncated at } \ell \leq L \right\}. \quad (24)$$

For $L = 1$ (a dipole-only Θ), $\mathcal{C}_{\text{Teff}}^{(1)}$ collapses to a one-parameter family, which is why dipole-only Teff models can look superficially similar to two-moment closures. The moment we allow $L \geq 2$, the map $(T_{A_1}, T_{A_2}, \dots) \mapsto (\mathbf{f}, \chi)$ is not one-dimensional, and $\mathcal{C}_{\text{Teff}}^{(L)}$ becomes a closure region. This distinction is not cosmetic: it is the practical way Teff multipoles represent multi-beam or non-convex angular structure that is invisible to $(\mathcal{E}, \mathcal{F}^a)$ data alone.

The realizability constraint $\Theta^{(L)}(e) > 0$ for all e appearing in (24) bounds the parameter space of admissible Teff multipoles. At dipole level ($L = 1$, axisymmetric),

the constraint reduces to $|A| < 1$ where A is the dipole amplitude. At quadrupole level ($L = 2$, axisymmetric with $\Theta(\mu) = 1 + A\mu + Q(\mu^2 - \frac{1}{3})$), the admissible region in (A, Q) space is delimited by $\Theta_{\min}(\mu) > 0$, which is determined by the eigenvalue condition $\lambda_{\min}(T_{ab}) > -1$ on the traceless quadrupole tensor (see Sec. VID for the general statement and Fig. 2 for a visualization). This realizability bound is tested to machine precision in the verification suite.

Figure 3 previews both aspects: the induced high- ℓ tail of standard multipoles and the resulting closure region in a representative moment plane. In Secs. V and VI we add the two missing ingredients that make this representation a controlled reduced model: a tangency diagnostic that decides when Teff-only must be enriched by energy degrees of freedom, and realizability bounds enforcing $\Theta^{(L)}(e) > 0$.

Two structural properties of the induced tail follow by elementary angular algebra and are used in the sequel.

Proposition 1 (Perturbative angular support bound). *Let $\Theta(e) = 1 + \theta(e)$ with $\theta(e) = \sum_{\ell=1}^L \theta_\ell(e)$, where each θ_ℓ has angular support at rank ℓ . Suppose $F(\Theta) = \sum_{n=0}^N a_n \theta^n$ is the order- N truncation of an analytic composition. Then the angular harmonic support of $F(\Theta)$ does not exceed rank NL :*

$$F(\Theta)|_{\ell > NL} = 0. \quad (25)$$

Proof. Each power θ^n is a sum of products of at most n angular harmonics, each of rank $\leq L$. The Clebsch–Gordan product of two harmonics at ranks ℓ_1 and ℓ_2 has support at $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$. By induction, θ^n has support at $\ell \leq nL$. Therefore the truncation $\sum_{n \leq N} a_n \theta^n$ has support at $\ell \leq NL$. $\square$

Proposition 2 (Even-sector parity). *If the active angular multipoles in $\Theta(e) = 1 + \sum_\ell \theta_\ell(e)$ are all of even rank (i.e. $\theta_\ell = 0$ for all odd ℓ), then any pointwise composition $F(\Theta(e))$ is parity-even, and all odd-rank angular observables vanish:*

$$\int d\Omega e^{A_\ell} F(\Theta(e)) = 0 \quad \text{for all odd } \ell. \quad (26)$$

In particular, a pure-quadrupole $\Theta(e) = 1 + T_{ab}e^a e^b$ generates no dipole, no octupole, and no odd-rank moment in any observable.

Proof. Under the parity map $e \mapsto -e$, each even-rank harmonic $\theta_\ell(e)$ is invariant, so $\Theta(-e) = \Theta(e)$ and hence $F(\Theta(-e)) = F(\Theta(e))$. For odd ℓ , $e^{A_\ell} \mapsto -e^{A_\ell}$, so the integrand is odd and the integral vanishes. $\square$

Proposition 1 implies that a finite perturbative order N introduces a hard angular cutoff artifact at $\ell = NL$; the full nonlinear composition produces the infinite tail discussed above. Proposition 2 constrains the interpretation of pure-quadrupole configurations: all angular flux-type observables are identically zero, not merely small.

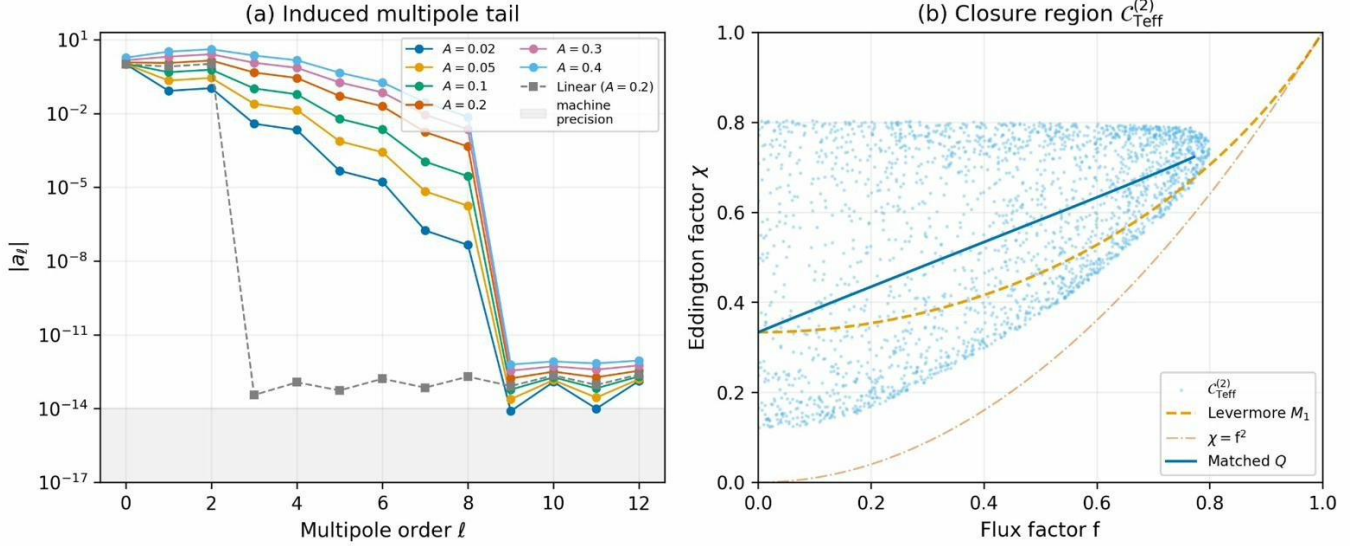


FIG. 3. Induced multipole tail and closure region for the $\ell \leq 2$ Teff truncation. (a) Legendre spectrum $|a_\ell|$ of the energy-density kernel $\rho(\mu) \propto \Theta(\mu)^4$ with $\Theta(\mu) = 1 + A\mu + Q(\mu^2 - \frac{1}{3})$ and $Q = A\sqrt{15}/4$ (RMS-matched), for dipole amplitudes $A = 0.02$ – 0.40 . Although Θ is truncated at $\ell = 2$, the nonlinear composition Θ^4 generates multipoles up to $\ell \approx 8$ – 9 before reaching the machine-precision floor (grey band). The dashed curve shows the linearised approximation $(1 + 4\theta)$ at $A = 0.20$: it captures $\ell \leq 2$ content but produces no tail beyond $\ell = 2$, confirming that the tail is a nonlinear effect (Eq. (22)). (b) Closure region $C_{\text{Teff}}^{(2)}$ (Eq. (24)) in the (f, χ) plane (teal scatter), obtained by sampling realizable (A, Q) pairs. The Levermore M1 curve (dashed orange) is a single-valued function $\chi(f)$; the Teff family spans a two-dimensional region with finite thickness $\Delta\chi$ visible already at $f \sim 0.2$. The matched- Q curve (blue solid) threads through the interior of the region. The dash-dotted pink line marks the realizability lower bound $\chi = f^2$.

D. Frame transformation of Teff multipoles

The spectral multipole tail of Sec. IIIB is a consequence of the nonlinear composition $e \mapsto \Phi(y/\Theta(e))$ at fixed observer. A second, kinematic, source of non-closure appears when the observer frame itself changes: a Lorentz boost generically maps an L -truncated Θ out of the L -truncated subspace. We record the exact law and its perturbative content here; the PSTF projection algebra underlying the perturbative coefficients is given in Appendix B3.

Notation. In this subsection only, v^a denotes the boost velocity (spatial, $v^a u_a = 0$); it should not be confused with the particle-speed ratio $v := \lambda/E$ of Sec. II. The weight exponent in Sec. IV is denoted p (not α) to avoid collision with the per-ray exponential-family intercept α of Sec. V.

Consider a boost from u^a to $\tilde{u}^a = \gamma(u^a + v^a)$ with $v^a u_a = 0$, $v^a v_a = \beta^2$, and Lorentz factor $\gamma = (1 - \beta^2)^{-1/2}$. For massless particles the Doppler factor is $D(e) = \gamma(1 + v_a e^a)$, and the boosted direction $\tilde{e}^a$ is obtained from e^a by the standard aberration map.

Theorem 3 (Exact Teff frame transformation, $m = 0$, $\mu = 0$). *For massless particles with vanishing chemical potential ($\mu = 0$) obeying $f_\Theta = \Phi(y/\Theta)$, the boosted an-*

gular deformation is

$$\tilde{\Theta}(\tilde{e}) = \frac{T_0}{\tilde{T}_0} \Theta(e(\tilde{e})) D(e(\tilde{e})), \quad (27)$$

where $e(\tilde{e})$ denotes the inverse aberration map and $\tilde{T}_0 = T_0 \langle \Theta D \rangle_\Omega$ restores the gauge $\langle \tilde{\Theta} \rangle_\Omega = 1$. The law is exact to all orders in β and holds identically for MB, BE, and FD statistics.

Proof. The occupation number $f(x, p)$ is a Lorentz scalar; hence $\Phi(y/\Theta) = \Phi(\tilde{y}/\tilde{\Theta})$. Because Φ is strictly monotone, $y/\Theta = \tilde{y}/\tilde{\Theta}$. With $\mu = 0$, the energy variable reduces to $y = E/(k_B T_0)$ and $\tilde{y} = \tilde{E}/(k_B \tilde{T}_0) = D E/(k_B \tilde{T}_0) = y D T_0/\tilde{T}_0$, which yields (27). $\square$

Proposition 4 ($\mu \neq 0$: one-field family is not Lorentz-closed). *For $\mu \neq 0$, the boosted shifted energy variable is*

$$\tilde{y} = \left(D \frac{T_0}{\tilde{T}_0} \right) y + (D - 1) \frac{\mu}{k_B \tilde{T}_0}, \quad (28)$$

where the second term is direction-dependent through the Doppler factor $D(e) = \gamma(1 + v_a e^a)$. The boosted distribution cannot generically be written in the pure one-field form $\Phi_\xi(\tilde{y}/\tilde{\Theta}(\tilde{e}))$; hence the one-field Teff family is not Lorentz-closed for $\mu \neq 0$. Closure under boosts requires

at least an affine two-parameter per-ray family with a direction-dependent additive shift.

Proof. Under a boost from u^a to $\tilde{u}^a$, the Lorentz scalar f satisfies $\Phi_\xi(y/\Theta) = \Phi_\xi(\tilde{y}/\tilde{\Theta})$ by frame invariance. Since Φ_ξ is strictly monotone, $y/\Theta = \tilde{y}/\tilde{\Theta}$. With $y = (E - \mu)/(k_B T_0)$ and $\tilde{E} = D E$, one obtains $\tilde{y} = (D E - \mu)/(k_B \tilde{T}_0) = D(T_0/\tilde{T}_0) y + (D - 1)\mu/(k_B \tilde{T}_0)$. The direction-dependent shift $(D - 1)\mu/(k_B \tilde{T}_0)$ cannot be absorbed into a multiplicative rescaling $\tilde{y} \mapsto \tilde{y}/\tilde{\Theta}(\tilde{e})$, since it is additive in $\tilde{y}$. $\square$

Since the tangency diagnostic (Sec. V) and entropy projection (Sec. VI) are evaluated in a fixed frame, this limitation does not affect the core results of Secs. V–VII.

The structure of (27) is transparent: Θ accounts for the original angular deformation, D encodes the Doppler blueshift/redshift, and $T_0/\tilde{T}_0$ is the gauge-restoring prefactor. None of these factors depends on the spectral shape Φ , so all statistics share the same transformation coefficients—a direct consequence of the factored form (16).

To extract the multipole content we expand (27) perturbatively in the boost velocity v^a and the existing Teff multipoles, project onto PSTF tensors, and collect terms order by order.

Proposition 5 (Perturbative Teff boost, covariant form, $\mu = 0$). *Under an infinitesimal boost v^a with $v^a u_a = 0$, $|v| \ll 1$, and with $\mu = 0$ (Theorem 3), and assuming the initial state is truncated at $\ell \leq 2$ ($T_{A_\ell} = 0$ for $\ell \geq 3$), the Teff parameters transform as*

$$\frac{\tilde{T}_0}{T_0} = 1 - \frac{1}{3} T_a v^a - \frac{1}{6} v_a v^a + \mathcal{O}(v^3), \quad (29)$$

$$\tilde{T}_a = T_a + v_a - \frac{4}{5} T_{ab} v^b + \mathcal{O}(v^2), \quad (30)$$

$$\tilde{T}_{ab} = T_{ab} + 2 v_{\langle a} T_{b \rangle} + v_{\langle a} v_{b \rangle} + \mathcal{O}(v^3), \quad (31)$$

$$\tilde{T}_{abc} = 3 T_{\langle ab} v_{c \rangle} + \mathcal{O}(v^2). \quad (32)$$

Each coupling coefficient has a geometric origin. The $-4/5$ in Eq. (30) arises because the product ΘD generates an $\ell = 1$ projection from the $\ell = 2$ quadrupole when contracted against the boost direction; the trace-free projection absorbs the factor $-8/15$ per axisymmetric component. The $+2$ in Eq. (31) is the standard dipole-velocity coupling that promotes the product $T_a v^b$ into its PSTF part. The octupole (32) is entirely induced: even if $T_{abc} = 0$ in the original frame, a nonzero quadrupole T_{ab} generates $\tilde{T}_{abc} = 3 T_{\langle ab} v_{c \rangle}$. An $L = 2$ truncation in one frame is therefore at best an $L = 3$ state in any other.

This coupling cascade,

$$\begin{aligned} \ell = 0 &\leftarrow T_a \cdot v, \\ \ell = 1 &\leftarrow v_a \oplus T_{ab} v^b, \\ \ell = 2 &\leftarrow v_{\langle a} T_{b \rangle} \oplus v_{\langle a} v_{b \rangle}, \\ \ell = 3 &\leftarrow T_{\langle ab} v_{c \rangle}, \end{aligned} \quad (33)$$

is the kinematic analog of the spectral multipole tail (Sec. III B). The spectral tail arises from the nonlinear energy composition y/Θ ; the boost-induced cascade arises from the nonlinear angular composition $\Theta \cdot D$. In both cases the finite-dimensional truncation is not preserved by the underlying symmetry, and the overflow encodes genuine physical content (spectral distortions and frame-dragging of the radiation pattern, respectively). Direct numerical evaluation of (27) confirms the perturbative coefficients to $\mathcal{O}(10^{-5})$ relative accuracy for $\beta \leq 0.01$.

Remark (Nonlinearity of Teff frame transformations). The standard energy-resolved PSTF multipoles $F_{A_\ell}(E)$ are tensor components of f on the mass shell; they transform linearly under boosts. The Teff multipoles T_{A_ℓ} parametrize a nonlinear submanifold of distribution space, so their boost law inherits that nonlinearity. Applying a linear boost rule to T_{A_ℓ} —as one might naïvely do by analogy with F_{A_ℓ} —would miss the $-4/5$ dipole-quadrupole coupling, the $+2$ quadrupole mixing, and the induced octupole entirely. For massive particles the situation is worse: the Doppler factor acquires energy dependence through the mass-shell relation $\lambda/E = \sqrt{1 - m^2/E^2} < 1$, giving $D(e, E) = \gamma(1 + v_a e^a \cdot \lambda/E)$. The product $\Theta(e) \cdot D(e, E)$ can no longer be absorbed into a purely angular redefinition $\tilde{\Theta}(\tilde{e})$ at fixed spectral shape, because the energy-dependent factor generates a spectral distortion that is not representable within the per-ray equilibrium family. The tangency diagnostic (Sec. V) detects this violation automatically. To see the scaling, expand $\lambda/E = 1 - \frac{1}{2}(m/E)^2 + \dots$ for $m \ll E$; the leading correction to the massless Doppler factor is $\delta D = -\gamma v_a e^a \cdot (m/E)^2/2$, which contributes an $n = 2$ Laguerre mode of amplitude $c_2 \sim \beta(m/\langle E \rangle)^2$ (where $\langle E \rangle \sim k_B T_0$ is the thermal energy scale). Hence the diagnostic scalar scales as $\varepsilon \sim \beta m^2/\langle E \rangle^2$; in the ultra-relativistic limit $m/\langle E \rangle \rightarrow 0$ the violation vanishes and the exact massless boost law (27) is recovered. For supernova ν_μ with $m \lesssim 0.05$ eV and $\langle E \rangle \sim 15$ MeV, the ratio is $m/\langle E \rangle \sim 10^{-9}$, rendering the mass correction entirely negligible.

Before closing, we comment on positioning versus maximum-entropy closures. Although the per-ray equilibrium family (16) is an exponential-family restriction along each direction, our construction is not “another Eddington factor.” Classical MaxEnt closures fix a finite set of moments and obtain an algebraic closure by entropy maximization, with accompanying hyperbolicity and entropy-pair structure for the closed moment system. Here the modeling choice is different: we restrict to a ray-wise equilibrium shape, keep Teff multipoles as coordinates on that manifold, and supply a diagnostic-and-upgrade rule when Teff-only ceases to be self-consistent. The distinction from algebraic closures such as those of Minerbo [36], Levermore [12, 37], and Dubroca and Feugeas [14] is therefore structural rather than parametric.

IV. PROJECTED TEFF HIERARCHY WITH COLLISIONS

We start from the covariant kinetic equation (13) and the PSTF multipole expansion (14)–(15) established in Sec. II. The purpose of this section is to make precise what we mean by a *projected* Teff hierarchy: we fix the hierarchy-coherent definitions of the collision multipoles, state how collisions are inserted *on* the Teff-only manifold defined in Sec. III, and identify the mass-matrix structure that governs invertibility of the reduced system. Proof-level statements about conservation laws and entropy production are deferred to Sec. VI and Appendix A.

A. Energy-resolved PSTF hierarchy and kinematic coupling pattern

Projecting the kinetic equation (13) onto the ℓ th PSTF sector yields the exact energy-resolved hierarchy

$$\mathbf{S}_{A_\ell}[F](x, E) = C_{A_\ell}[f](x, E), \quad (34)$$

where $\mathbf{S}_{A_\ell}$ is the linear streaming/kinematic operator acting on the set $\{F_{A_{\ell'}}\}_{\ell' \geq 0}$ and $C_{A_\ell}[f]$ are the collision multipoles defined below.

For our purposes we only need the structural pieces that organize the angular bookkeeping:

$$\begin{aligned} \mathbf{S}_{A_\ell}[F] \equiv & E \dot{F}_{\langle A_\ell \rangle} + \lambda D_{\langle a_\ell} F_{A_{\ell-1} \rangle} + \frac{\ell+1}{2\ell+3} \lambda D^b F_{b A_\ell} \\ & + \left(\text{terms involving } \theta, a^a, \omega_{ab}, \sigma_{ab} \text{ and } \partial_E \right). \end{aligned} \quad (35)$$

The two displayed gradient terms in Eq. (35) couple neighboring ranks $\ell \leftrightarrow \ell \pm 1$: the first transports angular structure from rank $\ell - 1$ into rank ℓ via spatial gradients, while the second removes rank- $(\ell + 1)$ content by contraction. The remaining kinematic terms (6), although lengthy in full form, do not change this message: expansion and acceleration appear with E -derivatives, vorticity mixes multipoles at fixed ℓ through the usual rotation generator, and shear generates additional couplings $\ell \leftrightarrow \ell \pm 2$ [29, 34]. The tedious and mundane enumeration of all kinematic contributions adds no conceptual content beyond these structural statements; we therefore suppress it here and refer to [34] for the complete form.

B. Collision multipoles and collision moment sources

To keep the hierarchy RHS unambiguous, we distinguish two objects.

(*Level 1*) *Energy-resolved collision multipoles.* For any collision operator $C[f](x, E, e)$ we define its PSTF coef-

ficients at fixed E by the same projection used for F_{A_ℓ} :

$$\begin{aligned} C[f](x, E, e) &= \sum_{\ell=0}^{\infty} C_{A_\ell}[f](x, E) e^{A_\ell}, \\ C_{A_\ell}[f](x, E) &:= \frac{1}{\Delta_\ell} \int d\Omega C[f](x, E, e) e_{\langle A_\ell \rangle}. \end{aligned} \quad (36)$$

Equation (36) is the only definition we use for the energy-resolved collision multipoles. It is *hierarchy-coherent* by construction: it matches the angular projection used on the streaming side.

(*Level 2*) *Collision sources for moment equations.* In applications we rarely evolve the full energy-resolved hierarchy (34); instead, we evolve a chosen set of weighted energy moments of Eq. (34). If the left-hand side is integrated with a weight $w(E, \lambda)$ (which is part of the model definition), then the corresponding collision source is the same weighted integral of $C_{A_\ell}(E)$:

$$\mathcal{C}_{A_\ell}^{[w]}[f] := \Delta_\ell \int_0^\infty d\lambda \lambda^2 w(E, \lambda) C_{A_\ell}[f](x, E). \quad (37)$$

Whenever we refer to “the collision multipole on the RHS of the ℓ th equation,” we must specify whether we mean $C_{A_\ell}(E)$ or the weighted source $\mathcal{C}_{A_\ell}^{[w]}$ for the particular hierarchy being evolved [32, 34].

Collision invariants constrain the low-rank sources: for channels that conserve particle number and/or energy-momentum, the corresponding collision moments vanish. In particular, the $\ell = 0$ source cannot be assigned an arbitrary relaxation ansatz without violating matching and conservation constraints; we treat this systematically in Sec. VI.

C. UU collision operator: formal insertion

Assumptions (UU collisions). When we specialize to quantum $2 \leftrightarrow 2$ interactions, we take $C = C^{\text{UU}}$ in Eq. (13) with the standard Uehling–Uhlenbeck gain-loss form

$$\begin{aligned} C^{\text{UU}}[f](p_1) &= \frac{1}{S} \int dP_2 dP_3 dP_4 W_{12 \rightarrow 34} \\ &\quad \times \left[f_3 f_4 (1 + \xi f_1) (1 + \xi f_2) \right. \\ &\quad \left. - f_1 f_2 (1 + \xi f_3) (1 + \xi f_4) \right], \end{aligned} \quad (38)$$

where $\xi = +1$ for bosons and $\xi = -1$ for fermions, S is a symmetry factor, and $W_{12 \rightarrow 34} \geq 0$ contains the matrix element together with the energy-momentum delta function $\delta^{(4)}(p_1 + p_2 - p_3 - p_4)$. Microreversibility is encoded by $W_{12 \rightarrow 34} = W_{34 \rightarrow 12}$ (up to exchange symmetries). Collision invariants are enforced by the same delta function and by the choice of interaction channel (number conserving versus number changing). These assumptions are the inputs

needed for the UU H -theorem and for the entropy statements we prove later [31, 38, 39].

D. Projection to the Teff manifold

We now restrict the distribution to the per-ray equilibrium family $f_\Theta = \Phi(y/\Theta)$ of Eq. (16), with Θ truncated at multipole order L as in Eq. (19) and the gauge $\langle \Theta \rangle_\Omega = 1$ (17). The reduced state is the finite collection $\{T_0, \mu, T_{A_\ell}\}_{1 \leq \ell \leq L}$.

A projected Teff hierarchy is obtained by applying a *parameter-space projection* to the reference hierarchy (34): rather than evolving the full set of energy-resolved multipoles $F_{A_\ell}(E)$, we evolve the Teff multipoles $T_{A_\ell}(x)$ and define all energy-resolved multipoles as derived functionals of (T_0, μ, T_{A_ℓ}) through Eq. (16). This fixes the streaming side uniquely once the $1 + 3$ kinematics are fixed; the remaining modeling choice is how to represent collisions *on* the Teff manifold in a way that is consistent with Eq. (36) and with the monopole matching constraints carried by (T_0, μ) .

Operationally, we evaluate the chosen collision operator at f_Θ and then project: $C_{A_\ell}[f_\Theta](E)$ is defined by Eq. (36) with $f = f_\Theta$. The induced collision structure is the input to the tangency diagnostic in Sec. V. For the reduced Teff evolution itself, we work with weighted moment hierarchies: for any weight $w(E, \lambda)$ and for $\ell \leq L$, we consider evolution equations of the schematic form

$$\Delta_\ell \int_0^\infty d\lambda \lambda^2 w(E, \lambda) S_{A_\ell}[F_\Theta](x, E) = \mathcal{C}_{A_\ell}^{[w]}[f_\Theta](x), \quad (39)$$

where F_Θ denotes the energy-resolved multipoles induced by f_Θ through Eq. (14). Equation (39) is a closed system once we specify the particular weights and the set of ℓ retained.

The chain rule on the Teff manifold converts the left-hand side of (39) into a mass-matrix form: since $F_{A_\ell}(E)$ depends on $T_{B_{\ell'}}$ through the nonlinear map $\Theta \mapsto \Phi(y/\Theta)$, the time derivative $\dot{F}_{A_\ell}$ involves the Jacobian $\partial F_{A_\ell} / \partial T_{B_{\ell'}}$. Integrating against $w(E, \lambda)$ produces a mass matrix

$$M_{A_\ell B_{\ell'}}^{[w]} := \Delta_\ell \int_0^\infty d\lambda \lambda^2 w(E, \lambda) \frac{\partial F_{A_\ell}}{\partial T_{B_{\ell'}}}, \quad (40)$$

so that the projected system takes the form $M_{A_\ell B_{\ell'}}^{[w]} \dot{T}_{B_{\ell'}} = (\text{spatial gradients}) + \mathcal{C}_{A_\ell}^{[w]}$. The mass matrix M is generically dense (coupling $\ell \leftrightarrow \ell'$ through Gaunt integrals), asymmetric (the PSTF normalization factor $1/\Delta_\ell$ sits on the row index only), and nonlinear in the Teff multipoles. For MB statistics at vanishing anisotropy with $w = y^p/(\lambda^2 E)$, the diagonal entries reduce to $\Gamma(p+2)\delta_{\ell\ell'}$; at $p = 3$ this gives $\text{diag}(24, 16)$ for the $L = 2$ axisymmetric case. For the factorised weight choices used in the present projected hierarchy, the symmetrised mass matrix is a positive weighted Gram matrix in angle:

the quadratic form $V^I M_{IJ}^{(s)} V^J = \int d\Omega q \Theta^p (\sum_I V^I \Psi_I)^2$ is strictly positive for any nonzero coefficient vector V whenever the angular basis functions $\{\Psi_I\}$ are linearly independent. In that setting, strict positivity $\Theta(e) > 0$ implies coercivity and hence invertibility of M ; the smallest eigenvalue scales as $\inf_e \Theta(e)^p$ (which equals $\Theta_{\min}^p$ for $p \geq 0$ and $\Theta_{\max}^p$ for $p < 0$). For more general weight choices $w(E, \lambda)$, positivity of Θ alone does not suffice, and invertibility must be checked at the level of the induced quadratic form (Proposition 16, Appendix B4).

Three comments are worth isolating. First, even if $\Theta(e)$ is truncated at $\ell \leq L$, the induced standard multipoles $F_{A_\ell}(E)$ generically develop an infinite high- ℓ tail through the nonlinear map $T_{A_\ell} \mapsto f_\Theta$ (20). This is why Eq. (39) is a projection of the full kinetic dynamics rather than a linear moment closure. Second, the $\ell = 0$ sector is constrained by collision invariants and by the matching conditions carried by (T_0, μ) ; we therefore do not enforce a universal “relaxation time” model for C_0 . Third, when linearized near an isotropic equilibrium, rotationally invariant kernels can yield approximately diagonal angular relaxation for $\ell \geq 1$, but we will not rely on that approximation in theorem-level statements [40].

Remark (Exact versus polynomial hierarchy evaluation). The projected system (39) admits two implementation levels distinguished by how the nonlinear map $T_{A_\ell} \mapsto F_{A_\ell}(E)$ and the resulting mass matrix M are evaluated. At *Level I* (exact quadrature), the map and its energy-weighted moments are computed numerically at each time step; M is dense, E -dependent, and the induced multipole tail is captured to full angular resolution, at a per-cell cost of $\mathcal{O}(N_\Omega L^2)$. At *Level J* (polynomial truncation at order N in $\{T_{A_\ell}\}$), M reduces to a sparse polynomial in the Teff multipoles: $N = 1$ recovers the linearized hierarchy with diagonal M and $\mathcal{O}(L)$ cost, while $N = 2$ restores quadratic Gaunt-coupled corrections and the induced tail through $\ell \leq 2L$, at $\mathcal{O}(L^2)$ cost. The benchmarks in Sec. VII use Level I throughout to isolate representation quality from polynomial approximation error; Level J is the natural pathway toward real-time transport solvers where per-cell cost is the binding constraint.

Preview: diagnose $\rightarrow$ switch. Equations (39) define what we mean by *Teff-only* reduced transport with collisions: all nonequilibrium is pushed into the angular dependence of $\Theta(e)$, and collision sources are inserted by the same PSTF projection used by the streaming operator. This reduced system is not guaranteed to remain self-consistent under the full dynamics. In Sec. V we develop a tangency diagnostic that detects when the residual contains energy-shape components that cannot be represented within the Teff-only family (16), and we turn that diagnostic into an upgrade protocol (activate one or two energy modes, or switch to multigroup) whenever Teff-only ceases to be self-consistent.

Remark (Linearized characteristic speeds at isotropic equilibrium). Although we do not validate the projected

hierarchy as a PDE solver in this work, the linearized characteristic structure at isotropic equilibrium ($T_{A_\ell} = 0$, $\Theta = 1$) can be read off the first-order perturbation of Eq. (39). For the simplest nontrivial case— $L = 1$, planar symmetry with spatial coordinate x —the two dynamical fields are the monopole perturbation $\delta\hat{\rho} := \delta\rho/\rho_0$ (normalised by the background energy density $\rho_0 \propto \int dE E^3 F_0(E)$) and the Teff dipole perturbation δT_1 , both dimensionless. Linearizing (39) around the isotropic background and dropping collision terms yields the homogeneous system

$$\partial_t \begin{pmatrix} \delta\hat{\rho} \\ \delta T_1 \end{pmatrix} + \begin{pmatrix} 0 & cR_\xi \\ c/(3R_\xi) & 0 \end{pmatrix} \partial_x \begin{pmatrix} \delta\hat{\rho} \\ \delta T_1 \end{pmatrix} = 0, \quad (41)$$

where $R_\xi := I_{\xi,p+1}/I_{\xi,p}$ is the ratio of consecutive spectral integrals $I_{\xi,n} := \int_0^\infty y^n \Phi_\xi(y) dy$, provided both integrals are finite. For Bose–Einstein statistics ($\xi = +1$), $\Phi_{+1}(y) \sim 1/y$ as $y \rightarrow 0^+$, so $I_{+1,n}$ is finite only for $n > 0$; the ratio R_{+1} therefore requires $p > 0$. For MB and FD with $\mu \leq 0$, $I_{\xi,n}$ is finite for $p > -1$, and the condition is automatically satisfied. Under these finiteness assumptions, $R_\xi \in (0, \infty)$ and the ratio cancels in the product of the off-diagonal entries ($cR_\xi \cdot c/(3R_\xi) = c^2/3$). For Maxwell–Boltzmann statistics ($\Phi_0 = e^{-y}$, $I_{0,n} = n!$), $R_0 = p + 1$ and the characteristic speeds are $\lambda_\pm = \pm c/\sqrt{3}$, recovering the same isotropic linear signal speed as M_1 . Hence, under the stated finiteness conditions, the same characteristic speeds $\lambda_\pm = \pm c/\sqrt{3}$ hold for all three statistics; R_ξ affects only the eigenvectors (i.e. the relative scaling of $\delta\hat{\rho}$ and δT_1), not the wave speeds. The system (41) is therefore strictly hyperbolic at the isotropic background for all three statistics, with real and distinct characteristic speeds bounded by the speed of light. (This is a local linearised characteristic preview, not a global nonlinear hyperbolicity theorem; the latter requires separate analysis deferred to future work.) Whether strict hyperbolicity persists for all realizable anisotropy amplitudes is an open question; the nonlinear energy-weighted angular integrals in M make the analysis substantially harder away from isotropy.

Scope of the present verification. Equation (39) defines a time-dependent PDE system for the Teff multipoles $T_{A_\ell}(x)$, coupled through nonlinear streaming and collision terms. In this paper we derive the projected hierarchy and verify its structural properties—collision-multipole coherence, entropy consistency (Sec. VI), and the tangency diagnostic (Sec. V)—but we do *not* evolve Eq. (39) as a spatial transport solver. The local-frame benchmarks (Sec. VII, Secs. VII.B–VII.E) test the diagnostic and representation mechanisms in the absence of streaming, and the slab benchmark (Sec. VII.F) uses a variable Eddington factor (VEF) iteration that evaluates the Teff closure quality without evolving the projected hierarchy directly.

Advancing the projected hierarchy as a self-contained PDE solver requires addressing several questions that lie outside the present scope: the characteristic structure

and hyperbolicity of the nonlinear system (39), which determines whether the Cauchy problem is well-posed; the compatibility of spatial discretization schemes (finite volume, discontinuous Galerkin) with the nonlinear Teff map and the realizability constraint $\Theta > 0$; and the behavior of the diagnose→switch protocol under the additional stiffness introduced by streaming–collision coupling. These are the central tasks of the planned follow-up work (Sec. VIII).

V. TANGENCY DIAGNOSTIC

The effective-temperature (Teff) representation is intentionally *Teff-only*: it captures nonequilibrium angular structure through a direction-dependent effective temperature $\Theta(x, e)$, while excluding independent energy-shape degrees of freedom at fixed direction. This exclusion is not a hidden approximation. It is a design choice that becomes trustworthy only when the kinetic dynamics is (approximately) tangent to the Teff manifold, and it becomes uncontrolled when collisions (or other physics) actively generate spectral distortions.

In this section we make this boundary explicit. We (i) state a sharp tangency criterion in energy space and (ii) turn it into a computable diagnostic, which we then use as a contract: we evolve the Teff-only hierarchy only while the diagnostic remains below tolerance, and we upgrade the representation when it does not. The diagnostic is evaluated from the same collision-operator evaluation that the hierarchy update already requires; the additional cost is a set of weighted energy integrals, not a second pass through the kinetics.

A. Normalized collision field and energy-mode expansion

Let us fix a spacetime point x and a direction e on the local momentum sphere. On the Teff manifold (Sec. III) the distribution takes the per-ray equilibrium form $f_\Theta(x, p) = \Phi(y/\Theta(x, e))$ with the dimensionless energy variable y defined in Eq. (16). Let $C[f]$ denote the collision operator evaluated on f . For the tangency question we only need the local-in- x structure of the right-hand side, so we introduce the *entropy-normalized collision field*

$$G_\xi(y, e) \equiv \frac{C[f_\Theta](y, e)}{f_\Theta(y, e)(1 + \xi f_\Theta(y, e))}, \quad (42)$$

viewed as a scalar field on $(y, e) \in (0, \infty) \times S^2$. For Maxwell–Boltzmann statistics ($\xi = 0$) this reduces to $G_0 = C/f_\Theta$; for quantum statistics the denominator $f(1 + \xi f) = 1/\eta''_\xi(f)$ is the inverse entropy Hessian (62) (whose second derivative is (59)), which provides the natural normalization connecting the tangent space of the per-ray manifold to affine functions of y (Appendix A 1).

Regardless of the statistics parameter ξ , it is convenient to isolate the energy-tangent content of G_ξ by an orthogonal expansion in generalized Laguerre polynomials, following the spirit of polynomial energy-space decompositions used in classical kinetic theory [41]. Let $L_n^{(-1)}(y)$ denote the boundary-case Laguerre family with the conventions $L_0^{(-1)}(y) = 1$ and $L_1^{(-1)}(y) = -y$. We use the weighted inner product

$$\langle g, h \rangle_* \equiv \int_0^\infty \frac{dy}{y} e^{-y} g(y) h(y), \quad (43)$$

for which the modes $n \geq 1$ satisfy the induced orthogonality relation

$$\int_0^\infty \frac{dy}{y} e^{-y} L_n^{(-1)}(y) L_m^{(-1)}(y) = \frac{1}{n} \delta_{nm}, \quad (n, m \geq 1). \quad (44)$$

The constant mode is not square-integrable under (43), but the modes $n \geq 1$ are well behaved, and the diagnostic below only uses the tail $n \geq 2$. This basis and weight are *independent of the statistics parameter ξ* ; the only quantity that changes between MB, BE, and FD is the normalization of G_ξ in (42).

We expand the entropy-normalized collision field in this basis as

$$G_\xi(y, e) = \sum_{n \geq 0} c_n^\xi(e) L_n^{(-1)}(y), \quad (45)$$

with coefficients defined by projection. For $n \geq 1$ one convenient normalization is

$$c_n^\xi(e) = n \int_0^\infty \frac{dy}{y} e^{-y} L_n^{(-1)}(y) G_\xi(y, e), \quad n \geq 1. \quad (46)$$

Caveat. Formula (46) extracts the true c_n only when the constant part c_0 of G_ξ vanishes. Because $L_0^{(-1)} = 1$ is not orthogonal to $L_n^{(-1)}$ for $n \geq 2$ under $\langle \cdot, \cdot \rangle_*$ (the cross inner product $\langle L_0, L_n \rangle_* = -1/n$ is finite), a nonzero c_0 contaminates the extracted c_n by an amount $-c_0$ (Lemma B.1). The regression-based diagnostic of Proposition 8(ii) is therefore the operationally correct extraction method.

Domain convention. Throughout this section, $y = (E - \mu)/(k_B T_0)$ with $\mu \leq 0$ (16), so that $y > 0$ for all physical momenta. This ensures the Laguerre inner product (43) on $(0, \infty)$ is well defined and the admissibility domain for Bose–Einstein statistics ($\Phi_{+1}(z)$ requires $z > 0$) is satisfied. For the degenerate extension $\mu > 0$, the Laguerre basis on $(0, \infty)$ would miss the negative- y domain. The natural repair is to recast the diagnostic in the strictly positive variable $x := E/(k_B T_0) > 0$, moving μ back into the exponential-family intercept (Sec. VIII C); the tangent space $\text{span}\{1, x\} = \text{span}\{1, y\}$ is unchanged.

B. Tangency criterion and interpretation

The tangency condition has a geometric origin. On each ray, the Teff ansatz $f = \Phi(y/\Theta)$ with $y = (E -$

$\mu)/(k_B T_0)$ is a one-parameter family in $\beta = 1/\Theta(e)$ with intercept $\alpha = 0$ (because μ is already absorbed into y). The per-ray manifold $\mathcal{M}_{\text{ray}}$ of Theorem 6 *enlarges* this to the two-parameter family $f_\xi(y; \alpha, \beta) = 1/(\exp(\alpha + \beta y) - \xi)$ by allowing $\alpha \neq 0$. The distribution-space tangent vectors are $\partial_\alpha f_\xi = -f_\xi(1 + \xi f_\xi)$ and $\partial_\beta f_\xi = -y f_\xi(1 + \xi f_\xi)$; after entropy normalization by $\eta_\xi'' = 1/[f_\xi(1 + \xi f_\xi)]$, these become -1 (constant) and $-y$ (linear). The entropy-normalized collision field $G_\xi := C[f_\xi]/[f_\xi(1 + \xi f_\xi)]$ therefore lies in $\text{span}\{1, y\}$ if and only if the collision forcing is tangent to $\mathcal{M}_{\text{ray}}$ at f_ξ . Since the actual Teff state has $\alpha = 0$ at every ray (because μ is absorbed into y), per-ray $\mathcal{M}_{\text{ray}}$ tangency is necessary but not sufficient for full Teff-manifold tangency; the additional angular-consistency condition on $c_0(e)$ is discussed in Remark V B and Proposition 7(e). The following result, proved in Appendix A 1, establishes the precise correspondence between per-ray manifold tangency and the vanishing of higher Laguerre modes in the entropy-normalized collision field.

Theorem 6 (Per-ray tangency criterion, all statistics ($\mu \leq 0$)). *Let the following hypotheses hold.*

(H1) (Statistics.) *The statistics parameter is $\xi \in \{-1, 0, +1\}$ (12), corresponding to Fermi–Dirac, Maxwell–Boltzmann, and Bose–Einstein statistics, respectively.*

(H2) (Per-ray equilibrium manifold.) *Fix a direction e in the local rest space of u^a . The per-ray equilibrium manifold $\mathcal{M}_{\text{ray}}$ at e is the two-parameter family*

$$f_\xi(y; \alpha, \beta) = \frac{1}{\exp(\alpha + \beta y) - \xi}, \quad \alpha \in \mathbb{R}, \beta > 0, \quad (47)$$

with the dimensionless energy variable y of Eq. (16). For $\xi = 0$, Eq. (47) reduces to the Maxwell–Boltzmann family $f_0(y) = \exp(-\alpha - \beta y)$.

(H3) (Admissible domain.) *On the energy domain $y \in (0, \infty)$, the distribution satisfies $f_\xi(y) > 0$; for fermions ($\xi = -1$) additionally $f_\xi(y) < 1$. (The latter is automatic from the form (47), but we state it explicitly because the proof divides by $f_\xi(1 + \xi f_\xi)$ and requires this product to be strictly positive. The domain $y > 0$ is guaranteed by the convention $\mu \leq 0$ fixed in Eq. (16).)*

(H4) (Collision operator.) *The collision operator $C[f_\xi]$ is a locally defined functional of f that produces a function $C[f_\xi](\cdot, e)$ on $(0, \infty)$ for each direction e .*

(H5) (Entropy-normalized field and regularity.) *The entropy-normalized collision field G_ξ of Eq. (42) satisfies $G_\xi = \eta_\xi''(f_\xi) \cdot C[f_\xi]$ where the factor $\eta_\xi''(f) = 1/[f(1 + \xi f)]$ is the entropy Hessian (62). We require that, for each direction e , the restriction $G_\xi(\cdot, e)$ belongs to the weighted Hilbert space*

$$\mathcal{H}_* := \{g : (0, \infty) \rightarrow \mathbb{R} \mid \|g - \bar{g}\|_* < \infty, \bar{g} \in \mathbb{R}\}, \quad (48)$$

where $\|u\|_*^2 = \langle u, u \rangle_*$ is the norm induced by (43). (The subtraction of a constant $\bar{g}$ accommodates the non-square-integrability of $L_0^{(-1)} = 1$ under the weight $y^{-1}e^{-y}$; the modes $n \geq 1$ are automatically in $L^2(y^{-1}e^{-y}dy)$ by the decay of e^{-y} and the polynomial growth of $L_n^{(-1)}$.) This regularity is satisfied whenever $G_\xi(\cdot, e)$ has at most polynomial growth as $y \rightarrow \infty$ and is locally bounded near $y = 0$ —conditions met by all collision operators considered in this work and, more generally, by any collision functional producing $C[f_\xi]$ of the same polynomial-exponential class as the equilibrium distribution itself. Under this regularity, the field admits a decomposition

$$G_\xi(y, e) = \sum_{n=0}^{\infty} c_n^\xi(e) L_n^{(-1)}(y) \quad (49)$$

in the generalized Laguerre basis of Sec. VA, with coefficients c_n^ξ for $n \geq 1$ uniquely determined by the mutual orthogonality (44) of $\{L_n^{(-1)}\}_{n \geq 1}$. (Because $L_0^{(-1)} = 1$ is not orthogonal to $L_n^{(-1)}$ for $n \geq 2$, the direct inner-product formula (46) extracts the true c_n only after the constant part c_0 is removed; see the caveat following (46).)

Then the collision forcing $C[f_\xi]$ is tangent to the per-ray manifold $\mathcal{M}_{\text{ray}}$ at f_ξ if and only if

$$c_n^\xi(e) = 0 \quad \forall n \geq 2, \forall e \quad (50)$$

Equivalently, tangency holds if and only if $G_\xi(\cdot, e)$ is an affine function of y for each direction e .

Remark (Why two parameters per ray; necessary vs. sufficient). The per-ray manifold $\mathcal{M}_{\text{ray}}$ in (H2) has two parameters (α, β) , while the Teff ansatz $f_\xi(y; \eta) = \Phi(y/\Theta)$ of Sec. VIA has one parameter $\eta = 1/\Theta$ per direction with $\alpha = 0$ (since μ is already absorbed into $y = (E - \mu)/(k_B T_0)$). The α -direction in $\mathcal{M}_{\text{ray}}$ represents a per-ray amplitude/normalisation degree of freedom that the Teff ansatz does not carry. Correspondingly, the two entropy-normalized tangent directions of $\mathcal{M}_{\text{ray}}$ split as follows. The constant mode c_0 (entropy-normalized direction $\eta_\xi'' \partial_\alpha f = -1$) represents a normalisation mismatch at each ray; a direction-independent c_0 can be absorbed by global energy-density matching, but a direction-dependent $c_0(e)$ cannot (Sec. IV). The linear mode c_1 (entropy-normalized direction $\eta_\xi'' \partial_\beta f = -y$) is absorbed by the per-ray temperature update $\delta\Theta(e)$. Modes $n \geq 2$ lie outside both parameter families and constitute the off-manifold content that $\mathcal{D}_{\geq 2}$ detects.

The condition $c_{n \geq 2} = 0$ is therefore *necessary* for full Teff-manifold tangency (if higher energy modes are present, no state update can absorb them), but not by itself *sufficient*: the constant-mode coefficient $c_0(e)$ must additionally be direction-independent (not an arbitrary direction-dependent intercept), so that the per-ray projections can be re-expressed as a single global Teff state.

This angular-consistency condition is enforced by the projected hierarchy (Sec. IV), not by the energy-mode diagnostic $\mathcal{D}_{\geq 2}$. In the controlled operators tested in Sec. VII (multipole relaxation, BGK, Thomson), $c_0(e)$ automatically satisfies the angular constraint because the collision operators respect the underlying symmetry.

Proposition 7 (Scope of the tangency criterion). *The tangency condition (50) is a per-ray, instantaneous, collision-side statement. It detects that the collision forcing $C[f_\xi]$ at the current state lies in the tangent space of the per-ray equilibrium manifold $\mathcal{M}_{\text{ray}}$. The following five properties are not implied by the tangency condition, individually or collectively:*

- (a) (No global invariance.) *Global-in-time invariance of the Teff manifold under the coupled streaming-collision dynamics.*
- (b) (No finite-time error bounds.) *Quantitative bounds on the discrepancy $\|f_{\text{Teff}} - f_{\text{kin}}\|$ between the Teff-only approximate solution and the full kinetic solution over a finite time interval.*
- (c) (No perturbative stability.) *Stability of the tangency condition under perturbation of the initial data, the collision kernel, or the angular truncation order.*
- (d) (No reduced-dynamics entropy monotonicity.) *Entropy monotonicity for the Teff-only reduced dynamics. Theorem 9 bounds the reduction-map entropy cost; it does not establish that the entropy functional is monotone along an approximate Teff-only trajectory.*
- (e) (No angular consistency of c_0 .) *Compatibility of the direction-dependent constant-mode coefficient $c_0(e)$ with a direction-independent normalisation adjustment. The tangency condition tests energy-mode content ($n \geq 2$) at each ray independently; it does not verify that the affine residual $(c_0(e), c_1(e))$ is angularly consistent with the projected-hierarchy constraints of Sec. IV (Remark VB).*

Proof. Items (a)–(c) follow from the local character of Theorem 6: the biconditional holds pointwise in space and time, for the collision operator alone. Item (d) follows from the distinction between the e-projection map $\mathcal{R}$ (Sec. VIA) and the time-evolution operator. $\square$

What the tangency condition *does* provide is an intrinsic and computable criterion for spectral distortion generation in the collision sector: the normal component of $C[f_\xi]$ relative to $\mathcal{M}_{\text{ray}}$ is exactly the higher-mode content $\{c_{n \geq 2}^\xi\}$, and the diagnostic scalar $d_{\geq 2}(e)$ is its norm. The physical content of the affine equivalence is transparent: the two tangent directions $\partial_\alpha f_\xi$ and $\partial_\beta f_\xi$ correspond to a ray-wise normalisation shift and a ray-wise temperature rescaling. Hence G_ξ affine in y means that the collision operator, at the current state, can be absorbed entirely

into updates of these two per-ray thermodynamic parameters. The moment G_ξ ceases to be affine, collisions are generating spectral shape changes that no rescaling of (α, β) can accommodate.

Remark (Relation to the Gorban–Karlin defect of invariance). The tangency criterion specialises the defect-of-invariance framework of Gorban and Karlin [18, 19, 42] to the per-ray exponential family in radiation transport. The entropy normalisation $G_\xi = C/[f(1+\xi f)] = \eta_\xi'' \cdot C$ maps the tangent space to $\text{span}\{1, y\}$, so the Laguerre coefficients $c_{n \geq 2}^\xi(e)$ are the energy-mode components of the entropy-normalised collision-side Gorban–Karlin defect $\eta_\xi'' \cdot (1 - P_{\text{ray}}) \cdot C[f_\Theta]$. The diagnostic (51) is the $\langle \cdot, \cdot \rangle_*$ -norm of this defect. Two operational differences should be noted. First, the present diagnostic monitors only the collision component of the defect; the streaming component is absorbed into the projected hierarchy (Sec. IV). In spatially varying media the streaming defect requires evolving the projected hierarchy as a PDE solver, which is the subject of future work (a standalone PDE solver). Second, whereas the Gorban–Karlin defect drives iterative manifold correction, the present diagnostic triggers a qualitative model switch: energy-shape activation or multigroup (the cost advantage lies primarily in the response mechanism—model switch versus iterative refinement—rather than in the diagnostic evaluation itself, which is comparably inexpensive for low-dimensional manifolds in both frameworks).

Remark (Formal scope versus benchmark evidence). Theorem 6 holds for all three quantum statistics under hypotheses (H1)–(H5). For MB, all hypotheses are unconditionally satisfied. For FD with $\mu \leq 0$, (H5) holds whenever C decays as f at large y . For BE with $\mu \leq 0$, the infrared regularity near $y = 0$ ($f \sim \Theta/y \rightarrow \infty$) is ensured by the UU gain–loss cancellation: dividing $C^{\text{UU}}[f_1]$ by $f_1(1+f_1)$ removes the divergent f_1 factor from both gain and loss (Appendix A). The benchmarks (Sec. VII) verify the biconditional under one controlled collision operator per statistics; testing on microphysical operators is deferred to future work.

Three consequences of Theorem 6 deserve brief mention. First, the Laguerre basis and inner product are statistics-independent: the entropy normalisation $G_\xi = C/[f(1+\xi f)]$ absorbs all ξ -dependence, and the subsequent decomposition is universal. Second, the entropy normalisation is the unique choice (up to a direction-dependent factor) for which tangent collisions produce an affine G : the naive normalisation C/f reports false-positive higher modes for $\xi \neq 0$ because $(1+\xi f)$ depends nontrivially on y . Third, the per-ray diagnostic $d_{\geq 2}(e) = \min_{a,b} |G_\xi - a - by|_*$ vanishes if and only if tangency holds on that ray; the aggregated diagnostic $\mathcal{D}_{\geq 2}(x)$ vanishes if and only if tangency holds at every direction simultaneously.

Remark (Weight-independence of the tangency zero-set). The tangency biconditional— $\mathcal{D}_{\geq 2} = 0$ iff $G_\xi \in \text{span}\{1, y\}$ —is a statement about membership in a finite-

dimensional subspace. For any two positive weight functions $w_1, w_2 > 0$ a.e. on $(0, \infty)$, the orthogonal-projection residuals $|G - \Pi_V^{(w_a)} G|_{L^2(w_a)}$ ($a = 1, 2$) vanish simultaneously, since $G \in V$ is weight-independent. Therefore the zero/nonzero classification of the diagnostic is invariant under the choice of energy inner product; what changes is the numerical value of the nonzero residual. This invariance underlies the extension to the x -coordinate Fisher weight in Paper II, where the tangent space $\text{span}\{1, x\} = \text{span}\{1, y\}$ is preserved under the coordinate change $y \mapsto x = y + \eta$.

The coefficient extraction for the diagnostic requires care, because direct Laguerre projection is numerically unreliable. The regression-based method described in the following proposition is not an implementation convenience but a structural requirement: without it, the diagnostic produces false positives of order unity even for exactly tangent states.

Proposition 8 (Regression-exact diagnostic extraction). *Under the hypotheses of Theorem 6, the per-ray diagnostic $d_{\geq 2}(e)$ can be computed by either of two methods, with qualitatively different numerical properties.*

- (i) (Direct Laguerre projection — structural L_0 leakage.) *The inner-product formula (46) does not yield the true c_n when the constant part c_0 is nonzero, because $\langle L_0, L_n \rangle_* = -1/n$ is finite and nonzero for $n \geq 2$ (Lemma B.1). Concretely, an exactly affine $G = a + by$ produces $c_2^{\text{extracted}} = -a$ via Eq. (46), even though the true $c_2 = 0$. This is an exact algebraic contamination, not a truncation artifact.*
- (ii) (Weighted affine regression — leakage elimination.) *Define $d_{\geq 2}^{\text{reg}}(e) := \min_{a,b \in \mathbb{R}} \|G_\xi(\cdot, e) - a - by\|_*$. The L_0 artifact is eliminated identically; for MB under a tangent collision, $d_{\geq 2}^{\text{reg}}$ reaches machine precision ($\lesssim 10^{-15}$).*
- (iii) (Equivalence for well-resolved modes.) *When $c_{n \geq 2}$ are well-separated from the leakage, the regression recovers the same $d_{\geq 2}$ as perfect direct projection, to within quadrature error.*

Proof. Lemma B.1 in Appendix B for (i); standard weighted least-squares for (ii); orthogonal-subspace identity for (iii). $\square$

C. A scalar diagnostic and an auditable upgrade trigger

The exact tangency condition (50) is too strict for numerical work: a pointwise-in-direction biconditional cannot be monitored at every timestep in a production code. To this end, we scalarize the higher-mode content into a single diagnostic that measures how strongly the collision operator attempts to leave the Teff-only manifold in energy space.

The per-ray diagnostic is defined as the weighted affine regression residual

$$d_{\geq 2}(e) \equiv \min_{a, b \in \mathbb{R}} |G_\xi(\cdot, e) - a - by|_*, \quad |u|_*^2 \equiv \langle u, u \rangle_* \quad (51)$$

This is well defined even though G_ξ may have a nonzero constant part c_0 with $\|c_0 \cdot 1\|_* = \infty$ (the constant function is not square-integrable under $\langle \cdot, \cdot \rangle_*$): the regression subtracts the affine part before measuring the residual norm. By the mutual orthogonality (44) of $\{L_n^{(-1)}\}_{n \geq 1}$ and the leakage-elimination property of regression (Proposition 8(ii)),

$$d_{\geq 2}(e)^2 = \sum_{n \geq 2} \frac{|c_n^\xi(e)|^2}{n}, \quad (52)$$

where the c_n ($n \geq 2$) are the true Laguerre coefficients—equivalently, those returned by the regression of Proposition 8(ii)—or, in practice, a truncated sum up to $n \leq N_{\max}$.

To obtain a single scalar at each spacetime point we aggregate over angles; for example, the $L^2(S^2)$ diagnostic

$$\mathcal{D}_{\geq 2}(x) \equiv \left(\int d\Omega d_{\geq 2}(x, e)^2 \right)^{1/2}, \quad (53)$$

or the stricter $L^\infty(S^2)$ proxy $\mathcal{D}_{\geq 2}^\infty(x) \equiv \sup_e d_{\geq 2}(x, e)$. From now on we treat “collision-side tangency holds” as the falsifiable condition

$$\mathcal{D}_{\geq 2}(x) \leq \varepsilon_{\text{tol}}, \quad (54)$$

with a tolerance ε_{tol} calibrated to the intended accuracy of the transport update and to the chosen numerical approximation of G_ξ . The tolerance is not a tuning parameter in the usual sense: it is chosen to match the truncation error of the spatial discretization (so that the diagnostic does not trigger upgrades for spectral distortions smaller than the scheme error), and its value is reported alongside all numerical results. The spirit here is analogous to realizability monitoring in entropy-based closures [12, 16], except that our diagnostic tests manifold tangency rather than moment realizability.

Remark (Two distinct diagnostic quantities). The tangency diagnostic $\mathcal{D}_{\geq 2}$ requires evaluating the collision operator $C[f_\Theta]$ and projecting the entropy-normalized field G_ξ onto the Laguerre basis in energy space. It tests *dynamical self-consistency*: whether collisions attempt to push the distribution off the Teff manifold. A separate quantity arises in settings where an independently computed angular distribution $f_{\text{ref}}(\mu)$ (e.g., from an S_N solve) is available: the angular projection residual

$$\mathcal{R}_{\text{ang}}(x) \equiv \frac{\|f_{\text{ref}}(\cdot) - f_{\Theta^*}(\cdot)\|_{L^2(S^2)}}{\|f_{\text{ref}}(\cdot)\|_{L^2(S^2)}}, \quad (55)$$

where Θ^* denotes the best-fit Teff parameters obtained by moment matching against f_{ref} . This measures *representation quality*: how well the Teff ansatz approximates a given angular distribution, irrespective of the

collision dynamics that produced it. The two diagnostics are correlated—large $\mathcal{R}_{\text{ang}}$ implies angular structure outside the Teff family, which generically produces large $\mathcal{D}_{\geq 2}$ once collisions act on the mismatch—but they are not identical. In a slab-geometry benchmark with a reference S_N solution, $\mathcal{R}_{\text{ang}}$ measures how well the Teff ansatz approximates the angular distribution; the collision-based $\mathcal{D}_{\geq 2}$ is exercised in the controlled-operator verifications of Sec. VII.

Upgrade protocol. When $\mathcal{D}_{\geq 2}(x)$ exceeds the prescribed tolerance, we activate the leading energy-shape mode if the $n = 2$ coefficient dominates the diagnostic norm (one-mode upgrade), or switch to multigroup energy discretisation if multiple modes contribute comparably. The operational protocol—including threshold calibration, hysteresis for switching stability, computational cost estimates, and phase-space positivity bounds for the upgraded state—is deferred to a companion solver paper.

D. Thomson scattering: an analytical tangency test

Thomson scattering provides a canonical analytical test because it isotropises without transferring energy.

At linear order in the anisotropy $\theta(e) = \Theta(e) - 1$, the collision field $C = -\theta y f(1 + \xi f)/\tau + \mathcal{O}(\theta^2)$ gives

$$G_\xi(y, e) = -\frac{\theta(e)y}{\tau} + \mathcal{O}(\theta^2), \quad (56)$$

which is affine in y : the tangency diagnostic vanishes at linear order for all three statistics. The proof follows from the equilibrium identity $-\Phi'(z) = \Phi(z)(1 + \xi\Phi(z))$ and the gauge condition $\langle \theta \rangle_\Omega = 0$ (Appendix A). At second order, the isotropised distribution $\langle \Phi(y/\Theta) \rangle_\Omega$ is a mixture of equilibria at different temperatures—not itself an equilibrium—and the entropy-normalised field acquires a non-affine correction

$$G_\xi^{(2)} = \frac{\langle \theta^2 \rangle - \theta(e)^2}{2\tau} [y^2(1 + 2\xi\Phi(y)) - 2y], \quad (57)$$

whose y^2 content projects dominantly onto the $n = 2$ Laguerre mode. For MB ($\xi = 0$) the bracket reduces to $2L_2^{(-1)}(y) - 2L_1^{(-1)}(y)$ with no higher-mode leakage; for quantum statistics the factor $2\xi\Phi(y)$ couples to all $n \geq 2$. The diagnostic thus scales as $d_{\geq 2}^{\text{Thomson}} = \mathcal{O}(\theta^2/\tau)$, confirming that the diagnostic fires at the correct perturbative order.

E. Energy-mode fingerprint

The Laguerre coefficients $\{c_n\}$ classify the type of spectral distortion. Table I shows four representative departures for BE at $\hat{\mu} = 1$.

TABLE I. Laguerre energy-mode fingerprint of representative spectral distortions for Bose–Einstein statistics.

Type	c_2	c_3	c_4	$d_{\geq 2}$	Upgrade
Temperature	7×10^{-5}	-2×10^{-5}	-8×10^{-6}	5×10^{-5}	(tangent)
y -type	1.4×10^{-2}	-3.5×10^{-3}	-1.6×10^{-3}	9.9×10^{-3}	one-mode
i -type	7.3×10^{-3}	-1.4×10^{-3}	-4.7×10^{-4}	5.2×10^{-3}	one-mode
μ -type	9.4×10^{-4}	7.8×10^{-4}	6.5×10^{-4}	9.2×10^{-4}	multigroup

A c_2 -dominated fingerprint (y -type, i -type) calls for the one-mode upgrade; a broadly distributed pattern (μ -type, $|c_2| \approx |c_3| \approx |c_4|$) calls for multigroup decomposition. The mode structure determines not merely *whether* the upgrade fires but *which* upgrade is appropriate: the diagnostic is a spectral distortion classifier, not merely an alarm.

Illustrative verification. Figure 4 makes the content of Theorem 6 visually concrete. At a fixed forward direction ($\mu = 0.8$), we plot the entropy-normalised collision field $G_\xi(y)$ for two collision operators applied to the same dipole Teff state ($A = 0.15$). The Teff-multipole relaxation $\dot{T}_\ell = -T_\ell/\tau_\ell$ produces $G_\xi(y) = -y \dot{\Theta}/\Theta^2$ —exactly affine in y , a straight line—and the diagnostic vanishes to machine precision ($d_{\geq 2} \sim 10^{-16}$). BGK isotropisation $C = \langle f_\Theta \rangle_\Omega - f_\Theta$ produces a G that curves in y , because a mixture of per-ray equilibria at different temperatures is not an equilibrium; the higher Laguerre modes $c_{n \geq 2}$ detect this curvature ($d_{\geq 2} \sim 10^{-2}$). Panel (b) confirms that the non-tangent diagnostic scales as A^2 , while the tangent diagnostic stays at machine precision regardless of A .

VI. ENTROPY STRUCTURE AND REALIZABILITY CONSTRAINTS

The Teff representation restricts the distribution to a submanifold; the diagnostic of Sec. V monitors collision-side departure. This section quantifies the reduction-map entropy cost of the Teff restriction and establishes realizability certificates.

A. Per-ray exponential family and entropy Hessian

At fixed direction e , the per-ray equilibrium family (16) takes the standard exponential-family form

$$f_\xi(y; \eta) = \frac{1}{\exp(\eta y) - \xi}, \quad \eta := \frac{1}{\Theta(e)} > 0, \quad (58)$$

with natural parameter $\eta = 1/\Theta$ and sufficient statistic y [24, 26, 43]. (In the $y = (E - \mu)/(k_B T_0)$ coordinate the log-normalisation vanishes; the intercept α of (47) is identically zero for the Teff subfamily.) This is the one-parameter subfamily of the two-parameter manifold $\mathcal{M}_{\text{ray}}$ of Theorem 6 (Remark VB); the additional

constant-mode direction in the tangent space of $\mathcal{M}_{\text{ray}}$ represents a per-ray amplitude mismatch (Remark VB).

Two quantities govern both the diagnostic and the entropy gap. The per-mode Fisher information is $I_\xi(\eta) = f(1 + \xi f)$, whose reciprocal is the entropy Hessian

$$\eta''_\xi(f) = \frac{1}{f(1 + \xi f)}, \quad (59)$$

which has already appeared as the normalisation of the collision field G_ξ (Eq. (42)) and will reappear as the weight in the entropy gap (Theorem 9). That the same factor plays both roles—straightening the tangent space of $\mathcal{M}_{\text{ray}}$ to $\text{span}\{1, y\}$ in the diagnostic (Theorem 6, Sec. VB) and weighting the reduction-map information cost—is the structural consequence of the exponential-family setting [24, 44]. (The higher Laguerre coefficients $c_{n \geq 2}$ can be viewed as measuring the departure of the spectral curve from an exponential-family geodesic in the sense of [43]; we note this interpretation but do not develop it into a formal curvature theorem.)

The ray-wise reduction proceeds in two layers. The *per-ray projection*

$$f_{\text{ray}}^*(\cdot, e) := \arg \min_{g \in \mathcal{M}_{\text{ray}}} D_\xi(f(\cdot, e) \| g) \quad (60)$$

maps f onto the two-parameter per-ray manifold $\mathcal{M}_{\text{ray}}$ (47) at each direction. By the standard exponential-family result, this is equivalent to matching both the per-ray normalisation $\int f dy$ and the per-ray mean $\int y f dy$, i.e. the moments dual to the sufficient statistics $(1, y)$ of $\mathcal{M}_{\text{ray}}$. Because $\mathcal{M}_{\text{ray}}$ is two-dimensional, the e-projection absorbs modes $n = 0$ (normalisation) and $n = 1$ (mean energy); the residual $f - f_{\text{ray}}^*$ has entropy-normalised Laguerre content entirely in $n \geq 2$.

The *global Teff representative* $f_{T_{\text{eff}}}[f]$ is the member of $\mathcal{M}_{T_{\text{eff}}}$ (Sec. III, with shared (T_0, μ) and gauge $\langle \Theta \rangle_\Omega = 1$) obtained by re-expressing f_{ray}^* in the $(T_0, \mu, \Theta(e))$ parameterisation:

$$T_0 \Theta(e) := \frac{1}{\beta^*(e)} \cdot \frac{1}{k_B}, \quad T_0 := \langle T_0 \Theta \rangle_\Omega, \quad (61)$$

where $\beta^*(e)$ is the slope parameter of the per-ray projection (60). When f_{ray}^* has $\alpha^*(e) = 0$ for all e (which occurs whenever f is already a Teff state, since the Teff ansatz has $\alpha = 0$ in the y -coordinate), f_{ray}^* and $f_{T_{\text{eff}}}[f]$ coincide; in general $\alpha^*(e) \neq 0$ and they differ by an amplitude mismatch (Theorem 10). The result can be re-expressed in the $(T_0, \mu, \Theta(e))$ parameterisation of Sec. III by choosing $T_0 = [\rho[f_{T_{\text{eff}}}] / a_\xi]^{1/4}$ (or any other convenient gauge) and setting $\Theta(e)$ so that $T_0 \Theta(e)$ reproduces the per-ray effective temperature. (If f is already a Teff distribution, $f_{T_{\text{eff}}}[f] = f$ pointwise regardless of gauge choice, so $D_\xi(f \| f_{T_{\text{eff}}}[f]) = 0$.)

Product manifold vs. global Teff manifold. Since $\mathcal{R}$ projects independently at each direction, its image lies in the product manifold $\mathcal{M} := \prod_e \mathcal{M}_{\text{ray}}(e)$, which is

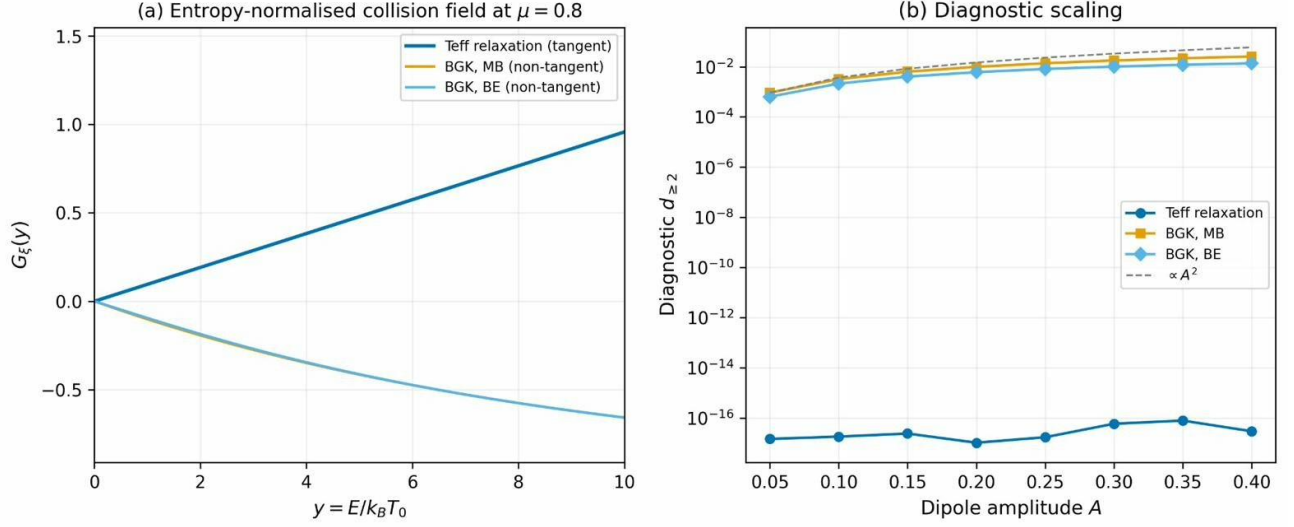


FIG. 4. Illustrative verification of the tangency criterion (Theorem 6). (a, top) Entropy-normalised collision field $G_\xi(y)$ at direction $\mu = 0.8$ and dipole amplitude $A = 0.15$: Teff-multipole relaxation (blue; exactly affine, $d_{\geq 2} \sim 10^{-16}$), BGK isotropisation at MB (vermillion; non-affine, $d_{\geq 2} = 0.017$), and BGK at BE (teal; non-affine, $d_{\geq 2} = 0.010$). (a, bottom) Residual $G - (a + by)$: the tangent collision produces a residual indistinguishable from zero; the BGK collisions produce oscillatory residuals. (b) Per-ray diagnostic $d_{\geq 2}$ versus dipole amplitude A : tangent collision (circles) at machine precision; BGK collisions (squares, diamonds) scale as $\propto A^2$.

generally larger than the global $\mathcal{M}_{T_{\text{eff}}}$ of Sec. III (the latter imposes a shared (T_0, μ) and the gauge $\langle \Theta \rangle_\Omega = 1$). Theorem 9 below is stated in terms of the ray-wise projection and holds on $\bar{\mathcal{M}}$; the restriction to $\mathcal{M}_{T_{\text{eff}}}$ is a stronger condition that the tangency diagnostic and projected hierarchy monitor separately (Proposition 7(e), Remark V B). The remaining energy-mode content ($c_{n \geq 2}$) lives in the same Laguerre sector that both the collision diagnostic $\mathcal{D}_{\geq 2}$ (53) and the state-residual diagnostic $\mathcal{D}_{\geq 2}^{\text{res}}$ (65) monitor (on different objects; see Theorem 11). By a standard result in information geometry [44, 45], this moment-matching map is simultaneously the e-projection (minimising $D_{\text{KL}}(f \| f_\Theta)$) and the maximum-entropy element at fixed $\langle y \rangle_f$. (The dual m-projection would instead match expectation parameters and minimise $D_{\text{KL}}(f_\Theta \| f)$; the direction of the KL divergence reverses, and the Pythagorean decomposition exchanges the two legs.)

B. Entropy normalisation and the H-theorem

The kinetic entropy density

$$\eta_\xi(f) := f \ln f - \xi^{-1}(1 + \xi f) \ln(1 + \xi f), \quad (62)$$

interpreting $\eta_0(f) = f \ln f - f$ by continuity at $\xi = 0$, induces the kinetic entropy four-current $s_{\text{kin}}^a = -k_B g / (2\pi\hbar)^3 \int dP p^a \eta_\xi(f)$. The standard UU H-theorem [31, 39] guarantees $\nabla_a s_{\text{kin}}^a \geq 0$, with equality iff f is a local equilibrium $f_{\text{eq}} = [\exp(\alpha + \beta_a p^a) - \xi]^{-1}$.

The Hessian (59) connects the H-theorem to the tangency diagnostic: dividing the collision operator by

$f(1 + \xi f)$ converts the tangent vectors into the constant functions $\{-1, -y\}$ (Theorem 6, proof Step 1), making the Laguerre decomposition statistics-independent.

C. Thermodynamic cost of the Teff reduction

The ray-wise reduction discards energy-shape content ($n \geq 2$ Laguerre modes). The following result bounds the entropy cost of the ray-wise reduction map. It is an application of the generalised Pythagorean theorem on dually flat manifolds [43, 44] to the per-ray family (58); the mathematical content is standard, and the contribution here is identifying its Laguerre-mode structure and its connection to the diagnostic.

Theorem 9 (Entropy-projection inequality). *Let $\xi \in \{-1, 0, +1\}$, let f be admissible ($f \geq 0$; for fermions, $0 \leq f \leq 1$ with $\mu \leq 0$), and let f_{ray}^* be the per-ray projection (60). Assume:*

- (H1) $\eta_\xi''(f) > 0$ on the physical domain. For fermions, both f and f_{ray}^* lie in the interior; this condition is automatic under $\mu \leq 0$.
- (H2) The moment-matching problem has a unique solution in $\mathcal{M}_{\text{ray}}$ for each e .

Then the reduced entropy exceeds the kinetic entropy,

$$S_{\text{kin}}[f] \leq S_{\text{red}}[f] := S_{\text{kin}}[f_{\text{ray}}^*], \quad (63)$$

with equality iff $f(\cdot, e) \in \mathcal{M}_{\text{ray}}$ for a.e. direction e (in particular, equality holds whenever $f \in \mathcal{M}_{T_{\text{eff}}}$, since

$\mathcal{M}_{T_{\text{eff}}} \subset \prod_e \mathcal{M}_{\text{ray}}$). The gap is the Bregman divergence

$$S_{\text{red}} - S_{\text{kin}} = k_B \int d\Pi D_\xi(f \| f_{\text{ray}}^*) \geq 0, \quad (64)$$

where $D_\xi(f \| g) := \eta_\xi(f) - \eta_\xi(g) - \eta'_\xi(g)(f - g)$.

The proof applies the Pythagorean theorem $D_\xi(f \| q) = D_\xi(f \| f^*) + D_\xi(f^* \| q)$ at the e-projection point $f^* = f_{\text{ray}}^*$ and is given in Appendix A.

The entropy gap of Theorem 9 can be bounded quantitatively by the state-residual diagnostic $\mathcal{D}_{\geq 2}^{\text{res}}$; the collision diagnostic $\mathcal{D}_{\geq 2}$ then predicts the rate at which this gap opens (Theorem 11).

Theorem 10 (Entropy-gap decomposition and bound). *Let $K \subset \mathcal{M}_{T_{\text{eff}}}$ be a compact subset on which $\Theta(e) \in [\Theta_{\min}, \Theta_{\max}]$ with $0 < \Theta_{\min} \leq \Theta_{\max} < \infty$. For any admissible f whose Teff projection $f_{T_{\text{eff}}}[f] \in K$ and whose per-ray projection $f_{\text{ray}}^*(\cdot, e)$ (matching both the amplitude and mean energy of f at each direction e) satisfies $|\alpha^*(e)| \leq \alpha_{\max}$ for some finite $\alpha_{\max}$, define the state-residual diagnostic*

$$\mathcal{D}_{\geq 2}^{\text{res}}[f] := \left(\int d\Omega \sum_{n \geq 2} \frac{[c_n^{(\text{res})}(e)]^2}{n} \right)^{1/2}, \quad (65)$$

where $c_n^{(\text{res})}(e)$ are the Laguerre coefficients of the entropy-normalized deviation $\eta''_\xi(f_{\text{ray}}^*) \cdot (f - f_{\text{ray}}^*)$ (the residual after removing the affine part by regression, Proposition 8). Then the Pythagorean theorem on $\mathcal{M}_{\text{ray}}$ gives

$$\frac{S_{\text{red}} - S_{\text{kin}}}{k_B} = \int d\Pi D_\xi(f \| f_{\text{ray}}^*) + \int d\Pi D_\xi(f_{\text{ray}}^* \| f_{T_{\text{eff}}}[f]), \quad (66)$$

where the first term is the $n \geq 2$ shape mismatch and the second is the amplitude mismatch. and the shape-mismatch term satisfies

$$\int d\Pi D_\xi(f \| f_{\text{ray}}^*) \leq C_2(K, \alpha_{\max}, \xi) [\mathcal{D}_{\geq 2}^{\text{res}}]^2. \quad (67)$$

The constant C_2 depends on K (controlling $\beta^* = 1/\Theta$) and $\alpha_{\max}$ (controlling the per-ray amplitude); together these give a uniform bound on $\eta''_\xi(f_{\text{ray}}^*)$. The amplitude-mismatch term is not captured by $\mathcal{D}_{\geq 2}^{\text{res}}$.

Proof. The Pythagorean identity for the per-ray Bregman divergence on the two-parameter family $\mathcal{M}_{\text{ray}}$ gives (66) [44]. For the first term: f_{ray}^* matches both the amplitude and $\langle y \rangle$ of f per ray, so the residual $f - f_{\text{ray}}^*$ has Laguerre content entirely in $n \geq 2$; a Taylor expansion of D_ξ around f_{ray}^* , with the compactness of K and the amplitude bound $|\alpha^*| \leq \alpha_{\max}$ jointly bounding $\eta''_\xi(f_{\text{ray}}^*)$ from above, yields (67). $\square$

Theorem 11 (Collision diagnostic as gap-opening rate predictor). *Let f_Θ be a state on the Teff manifold $\mathcal{M}_{T_{\text{eff}}}$, let $C[f_\Theta]$ be the collision operator evaluated on it, and assume the per-ray e-projection $f \mapsto f_{\text{ray}}^*$ is locally unique and differentiable in a neighbourhood of f_Θ .*

(a) *The collision-field diagnostic $\mathcal{D}_{\geq 2}$ (53) reads the $n \geq 2$ Laguerre content of the entropy-normalised collision field $G_\xi = C[f_\Theta]/[f_\Theta(1+\xi f_\Theta)]$. It does not measure the current energy-shape deviation of the state from the manifold: for states on $\mathcal{M}_{T_{\text{eff}}}$, the state-residual diagnostic satisfies $\mathcal{D}_{\geq 2}^{\text{res}} = 0$ identically, while $\mathcal{D}_{\geq 2}$ can be nonzero.*

(b) *After one collision step of duration Δt from a Teff state, the leading-order connection is*

$$\mathcal{D}_{\geq 2}^{\text{res}}(t+\Delta t) = \Delta t \mathcal{D}_{\geq 2}(t) + \mathcal{O}(\Delta t^2). \quad (68)$$

Hence $\mathcal{D}_{\geq 2}$ predicts the rate at which the shape-mismatch entropy gap opens, not the current gap itself.

Proof. Part (a): $\mathcal{D}_{\geq 2}^{\text{res}}$ measures the $n \geq 2$ energy-mode content of $f - f_{\text{ray}}^*$; on $\mathcal{M}_{T_{\text{eff}}}$ we have $f = f_{\text{ray}}^*$ (both modes $n = 0, 1$ are matched by construction), so $\mathcal{D}_{\geq 2}^{\text{res}} = 0$. Meanwhile $\mathcal{D}_{\geq 2}$ is computed from G_ξ , a different object. Part (b): a Taylor expansion of the evolved state $f(t+\Delta t) = f_\Theta(t) + \Delta t C[f_\Theta] + \mathcal{O}(\Delta t^2)$ gives $f - f_{\text{ray}}^* = \Delta t C[f_\Theta] + \mathcal{O}(\Delta t^2)$; dividing by $f(1+\xi f)$ and projecting onto $n \geq 2$ yields (68). $\square$

Remark (Operational consequence). In the diagnose-to-switch protocol, it is the collision diagnostic $\mathcal{D}_{\geq 2}$ that is monitored, since it can be evaluated at the current Teff state without stepping off the manifold.

Numerical calibration. On $K = \{|A| \leq 0.5, |Q| \leq 0.3\}$ with $L_2^{(-1)}$ spectral perturbations of amplitude $\varepsilon \in [10^{-4}, 10^{-1}]$ at isotropic background, the ratio $\Delta S / [\mathcal{D}_{\geq 2}^{\text{res}}]^2$ stabilises to a finite constant of order unity as $\varepsilon \rightarrow 0$ for MB (≈ 1.1) and FD (≈ 0.9), confirming the quadratic bound with a moderate prefactor. For BE, the infrared weight near $y = 0$ amplifies the Bregman integrand; the bound still holds but with a larger constant (~ 200), reflecting the stronger curvature of η_{+1} at high occupation.

Proposition 12 (Linearised quadratic law for shape mismatch). *For $\mathcal{D}_{\geq 2}^{\text{res}} \ll 1$, the shape-mismatch term in (66) satisfies*

$$\frac{1}{k_B} \int d\Pi D_\xi(f \| f_{\text{ray}}^*) = C_{\text{quad}} [\mathcal{D}_{\geq 2}^{\text{res}}]^2 + \mathcal{O}((\mathcal{D}_{\geq 2}^{\text{res}})^3), \quad (69)$$

with $C_{\text{quad}} = 1/2$ in the $\langle \cdot, \cdot \rangle_*$ -weighted norm.

Proof. Second-order expansion of $D_\xi(f \| f_{\text{ray}}^*)$; the residual $f - f_{\text{ray}}^*$ has Laguerre content entirely in $n \geq 2$ (both amplitude and $\langle y \rangle$ are matched by f_{ray}^*). $\square$

Corollary 13 (Reduction-map scope). *The above results are statements about the per-ray reduction $f \mapsto f_{\text{ray}}^*$ and the global Teff representative $f_{\text{Teff}}[f]$ at a single instant. They are not: (i) an H-theorem for the reduced dynamics; (ii) a bound on entropy production along trajectories; (iii) a monotonicity guarantee under time stepping. Each requires manifold invariance monitored by the tangency diagnostic (Proposition 7).*

A state-residual diagnostic value $\mathcal{D}_{\geq 2}^{\text{res}} \sim 10^{-3}$ implies a reduction-map entropy gap of order 10^{-6} (Theorem 10). The structural link between the collision diagnostic $\mathcal{D}_{\geq 2}$ and the entropy gap operates through $\mathcal{D}_{\geq 2}^{\text{res}}$ (Theorem 11): the $n \geq 2$ shape-mismatch part of the entropy gap (66) is bounded by $[\mathcal{D}_{\geq 2}^{\text{res}}]^2$. The amplitude-mismatch term $\int D_{\xi}(f_{\text{ray}}^* \| f_{\text{Teff}})$ is a separate contribution not captured by either diagnostic; for near-Teff states produced by time stepping, both terms scale as Δt^2 and the $n \geq 2$ part typically dominates.

Theorem 9 is a statement about the reduction map, not the time evolution. It does not imply monotonicity of S_{red} along an approximate Teff-only trajectory; entropy monotonicity for the reduced dynamics requires manifold invariance or an explicit projection step—precisely the condition the tangency diagnostic is designed to test.

We stress a point that is frequently confused in the closure literature: the MaxEnt construction defines only the reduction map (60) used to quantify information loss; we do not use entropy maximization to derive constitutive relations.

D. Realizability: positivity of a truncated $\Theta(e)$

The minimal physical requirement is

$$\Theta(e) > 0 \quad \text{for all directions } e, \quad (70)$$

since Φ_{ξ} is defined on a one-sided domain.

Proposition 14 (Pure-quadrupole realizability certificate). *Let $T_{ab} = T_{\langle ab \rangle}$ be a PSTF tensor and set $T_a = 0$ (no dipole). Define $\Theta(e) = 1 + T_{ab} e^a e^b$. Then $\Theta(e) > 0$ for all unit e iff $\lambda_{\min}(T) > -1$. Equivalently, $h_{ab} + T_{ab}$ is positive definite.*

Proposition 15 (Axisymmetric dipole+quadrupole realizability, sharp). *For $\Theta(\mu) = 1 + A\mu + Q(\mu^2 - \frac{1}{3})$ with $\mu \in [-1, 1]$, realizability $\Theta > 0$ holds iff (R1) $|A| < 1 + \frac{2}{3}Q$ and (R2) if $Q > 0$ and $|A| \leq 2Q$ then $3A^2 + 4Q^2 < 12Q$. The boundary is an ellipse $3A^2 + 4(Q - \frac{3}{2})^2 = 9$ joined to lines $|A| = 1 + \frac{2}{3}Q$ at C^1 continuity. At $A = 0$ this recovers Proposition 14.*

Proofs are in Appendix A. A sufficient algebraic condition for the general (non-axisymmetric) case is $|T_a| + \|T_{ab}\|_{\text{op}} < 1$ (Lemma B.2, Appendix B). The realizable region for the axisymmetric case is shown in Fig. 5.

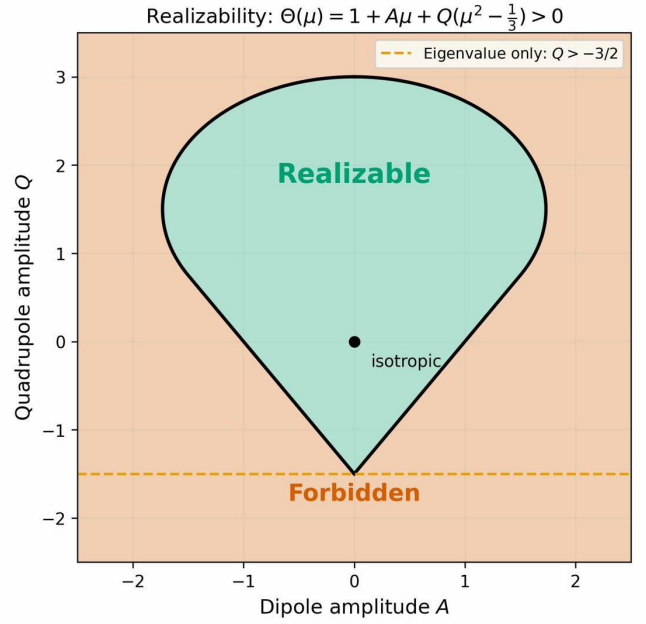


FIG. 5. Realizability boundary for the axisymmetric $\Theta(\mu) = 1 + A\mu + Q(\mu^2 - \frac{1}{3})$. Green: $\Theta_{\min} > 0$ (realizable); red: forbidden. The thick black curve is the exact $\Theta_{\min} = 0$ locus from Proposition 15. At $A = 0$ the boundary reduces to $Q \in (-3/2, 3)$, recovering the pure-quadrupole eigenvalue condition of Proposition 14; for $A \neq 0$ the dipole contracts the realizable domain below the eigenvalue-only prediction.

Two independent diagnostics. Realizability ($\Theta > 0$) is a hard constraint to be enforced at every step; the tangency diagnostic is a soft diagnostic signalling when spectral extensions are needed. The two operate independently. Propositions 14 and 15 are algebraic admissibility certificates for the parameter space of Teff multipoles; they characterise static admissibility, not invariant-region preservation under reduced evolution. Whether a time-stepping scheme for the projected hierarchy (39) maintains $\Theta(e) > 0$ under CFL-limited evolution is a numerical question deferred to a companion solver paper.

VII. STRUCTURAL VERIFICATION

The three figures accompanying the theory sections (Figs. 3, 4, and 5) verify the internal logic of the formal results: Fig. 3 confirms the induced multipole tail and closure-region structure predicted by the nonlinear Teff map; Fig. 4 makes the tangency biconditional of Theorem 6 visually concrete by contrasting a tangent and a non-tangent collision operator; and Fig. 5 displays the sharp realizability boundary of Propositions 14 and 15. All three use analytic or controlled-operator evaluations that exercise the representation and diagnostic mechanisms in isolation. The operators tested—multipole relaxation, BGK isoenergisation, Thomson scattering, and

a ν - e^- surrogate—are chosen for analytical tractability and mechanism coverage; they are not microphysical kernels. Generality to realistic neutrino or photon opacities remains untested and constitutes future work.

All benchmarks use the axisymmetric Teff ansatz truncated at $\ell \leq 2$,

$$\Theta(\mu) = 1 + A\mu + Q\left(\mu^2 - \frac{1}{3}\right), \quad (71)$$

with $\mu = \cos\theta \in [-1, 1]$, dipole amplitude A , and quadrupole amplitude Q . Angular quadrature is performed on a Gauss–Legendre grid; energy integrals use a log-spaced grid with weight $(dy/y)e^{-y}$ matched to the Laguerre inner product.

Tangency across quantum statistics. The tangency biconditional is tested by applying a temperature-multipole relaxation operator $dT_\ell/dt = -T_\ell/\tau_\ell$ to the ansatz (71) across all three statistics. We form the entropy-normalised collision field

$$G_\xi(y, \mu) := \frac{C[f_\Theta](y, \mu)}{f_\Theta(y, \mu)(1 + \xi f_\Theta(y, \mu))}, \quad (72)$$

expand in generalised Laguerre polynomials $L_n^{(-1)}(y)$, and extract the coefficients $c_n^\xi(\mu)$. For all three statistics—Maxwell–Boltzmann, Fermi–Dirac, and Bose–Einstein—the coefficients $c_{n \geq 2}$ vanish to machine precision ($d_{\geq 2} \lesssim 10^{-16}$) under multipole relaxation, confirming that Theorem 6 holds in the tangent direction for every $\xi \in \{-1, 0, +1\}$.

Under a BGK isoenergetic collision operator (non-tangent test), quantum nonlinearity ($1 \pm f$) in G_ξ generates non-trivial higher modes for Fermi–Dirac and Bose–Einstein. At leading order in the anisotropy amplitude, the second-order correction (57) predicts that the diagnostic ratio reduces to a statistics-dependent spectral integral:

$$\frac{d_{\geq 2}^{\text{BE}}}{d_{\geq 2}^{\text{FD}}} \approx \frac{\|y^2(1 + 2\Phi_{+1}) - 2y\|_*}{\|y^2(1 - 2\Phi_{-1}) - 2y\|_*} \approx 0.66, \quad (73)$$

where $\|\cdot\|_*$ denotes the $n \geq 2$ norm in the Laguerre inner product (43). This leading-order ratio is A -independent because the angular-variance prefactor $[\langle\theta^2\rangle - \theta(e)^2]$ in (57) cancels in the ratio. At finite amplitude ($A \gtrsim 0.1$), higher-order corrections beyond the $O(\theta^2)$ expansion introduce A -dependence: the measured ratio rises from ≈ 0.67 at $A = 0.05$ to ≈ 0.91 at $A = 0.30$. The structural message is that the diagnostic cleanly separates the statistics-dependent spectral content from the angular geometry, with the leading-order ratio set by a computable spectral integral and the finite-amplitude drift reflecting the nonlinear composition $\Phi(y/\Theta)$.

Nonlinear tail and closure region. The nonlinear Teff map $\Theta \mapsto \Theta^4$ generates a multipole tail extending to $\ell \sim 8$ –9 from the $\ell \leq 2$ input (Fig. 3a), and the resulting closure region $\mathcal{C}_{\text{Teff}}^{(2)}$ (Fig. 3b) is a two-dimensional

TABLE II. Evidence-calibration table: collision operator, statistics, diagnostic test, evidence level, and section reference.

Operator	ξ	Evidence	Sec.
Multipole relax.	MB	Machine- ε tangency	VII
Multipole relax.	FD	Machine- ε tangency	VII
Multipole relax.	BE	Machine- ε tangency	VII
BGK isoenergetic	MB	$d_{\geq 2} \propto A^2$	VII
BGK isoenergetic	FD	Non-tangent, A -dep. ratio	VII
BGK isoenergetic	BE	Non-tangent, A -dep. ratio	VII
Thomson	MB	Linear tangency, $O(A^2)$	VII
Thomson	FD	Linear tangency, $O(A^2)$	V
Thomson	BE	Linear tangency, $O(A^2)$	V
ν - e^- surr.	MB	Non-tangent (semi-real.)	VII

domain in $(\mathfrak{f}, \chi)$ space that extends beyond the M_1 Levermore curve in the beam-crossing regime. The linearised approximation $(1 + 4\theta)$ captures $\ell \leq 2$ content but produces no tail, confirming the nonlinear origin.

Tangent versus non-tangent collision. Figure 4 (Paper I) makes the content of Theorem 6 visually concrete. At a fixed forward direction ($\mu = 0.8$), the Teff-multipole relaxation produces G exactly affine in y ($d_{\geq 2} \sim 10^{-16}$); BGK isotropisation produces G with curvature detected by $c_{n \geq 2}$ ($d_{\geq 2} \sim 10^{-2}$). The non-tangent diagnostic scales as A^2 (Fig. 3b), confirming that the departure is quadratic in the anisotropy amplitude. This scaling is analytic: for MB statistics with $\Theta(\mu) = 1 + A\mu$ at small A , the entropy-normalized collision field $G(y) = -A\mu y + O(A^2)$ has $c_0, c_1 = O(A)$ (tangent-space updates to a_0, a_1), while $c_{n \geq 2}$ first appears at $O(A^2)$ because the isotropic-average operator $\bar{f}(y)$ preserves the per-ray spectral shape to linear order and generates the quadratic residual from the variance of Θ across directions. When $d_{\geq 2}$ exceeds a prescribed tolerance, the diagnostic signals that one energy-shape mode should be activated; threshold calibration, hysteresis, and cost analysis for this upgrade are deferred to future work.

Evidence-calibration summary. Table II summarises the evidence level for each collision operator tested.

Scope map. Table III separates results formally proved and verified within the present paper from those deferred to future work. No claim in the present paper depends on results from subsequent papers in the series.

Diagnostic scope. The collision diagnostic $\mathcal{D}_{\geq 2}$ is verified as an energy-mode tangency trigger: it produces zero false positives at isotropy ($A = 0$) and scales monotonically with anisotropy amplitude. However, $\mathcal{D}_{\geq 2}$ alone does not detect angular-truncation failures beyond the active L manifold ($\ell > L$ streaming content); an additional streaming-side proxy would be needed for that purpose.

Universal versus statistics-dependent properties. Table IV catalogues which formal results and geometric properties are universal across quantum statistics (within

TABLE III. Scope map: formal results proved here, numerically verified here, and deferred to future work.

Result	Where
<i>Formally proved (this paper)</i>	
Tangency iff (Thm. 6)	Here
Diagnostic semantics (Thm. 11)	Here
Entropy-projection (Thm. 9)	Here
Boost law (Thm. 3; Prop. 5)	Here
$\mu \neq 0$ non-closure (Prop. 4)	Here
Perturbative support (Prop. 1)	Here
Even-sector parity (Prop. 2)	Here
Realizability certificates	Here
Induced tail mechanism	Here
Conditional coercivity (Prop. 16)	Here
<i>Verified (controlled operators)</i>	
MB/FD/BE machine-precision tangency	Here
MB/FD/BE Thomson linear tangency	Here
Closure region (axi. $L=2$)	Here
<i>Not claimed</i>	
Global kinetic adequacy from $\mathcal{D}_{\geq 2}$ alone	—
Streaming-side failure detection	—
Dynamical realizability preservation	—
<i>Deferred to future work</i>	
Symmetric hyperbolicity	Future
Switching calibration	Future
VEF slab vs M_1	Future
Massive / degenerate extensions	Future
Realistic opacities	Future
Multi-species coupling	Future
Standalone PDE solver	Future

the present $\mu \leq 0$ domain construction) and which acquire quantitative ξ -dependence. The partition underscores the design: the diagnostic *mechanism* (Laguerre basis, regression extraction, threshold protocol) is statistics-independent; only the *amplitude* of the non-tangent residual depends on ξ .

TABLE IV. Universal (all ξ) versus statistics-dependent properties of the Teff representation, verified in the axisymmetric $L = 2$ sector. “Bench.” column: **F** formal theorem, **N** numerical.

Property	Univ.?	Bench.
Tangency biconditional (Thm. 6)	Yes	F
Laguerre basis $\{L_n^{(-1)}\}$	Yes	F
Regression diagnostic	Yes	F,N
Entropy-projection (Thm. 9)	Yes	F
Realizability certificates	Yes	F
Boost law	Yes	F
Induced tail spectrum	Yes	N
Closure-region geometry	Yes	N
Non-tangent residual $d_{\geq 2}$	No	N
BE/FD diagnostic ratio	No	N
Mass-matrix entries	No	F
Thomson $\mathcal{O}(\theta^2)$ prefactor	No	F
Fingerprint amplitudes	No	N

VIII. DISCUSSION AND OUTLOOK

The paper establishes three formal results and equips each with a physical interpretation. Theorem 6 provides a per-ray, instantaneous, collision-side criterion for when the Teff manifold absorbs collision forcing: the entropy-normalised field G_ξ must be affine in y , and the $n \geq 2$ Laguerre residual is the computable diagnostic. Theorem 9 bounds the thermodynamic cost of the ray-wise reduction by the $n \geq 2$ energy modes of the state residual $\mathcal{D}_{\geq 2}^{\text{res}}$ —an instance of the Pythagorean theorem on the dually flat per-ray exponential family [43, 44]. The collision diagnostic $\mathcal{D}_{\geq 2}$ lives in the same Laguerre sector but acts on a different object (G_ξ vs. the state deviation); it predicts the rate at which the entropy gap opens. The entropy gap decomposes (Theorem 10) into a shape-mismatch term controlled by the $n \geq 2$ modes of the state residual, and an amplitude-mismatch term not captured by $\mathcal{D}_{\geq 2}^{\text{res}}$. The collision diagnostic $\mathcal{D}_{\geq 2}$ predicts the rate at which the shape-mismatch gap opens (Theorem 11). The angular consistency of the per-ray amplitude sector ($\alpha^*(e) = 0$ vs. $\alpha^*(e) \neq 0$) is a separate condition (Remark VB, Proposition 7(e)). Propositions 14 and 15 provide sharp algebraic certificates for $\Theta(e) > 0$ in the pure-quadrupole and mixed dipole-plus-quadrupole sectors. The structural verification (Sec. VII) confirms the tangency biconditional—machine-precision tangency for all three statistics under multipole relaxation, quantified non-tangency for quantum statistics under BGK with a diagnostic ratio set by a computable spectral integral—and demonstrates the closure-region geometry predicted by the nonlinear Teff map. All three figures exercise the representation and diagnostic on controlled collision operators; quantitative benchmarks against reference transport solutions are deferred to future work.

A. Validity regime

The Teff ansatz holds when three structural properties of the collision kernel are satisfied: the fractional energy transfer is small ($\Delta E/E \ll 1$, enabling a Fokker–Planck reduction that produces at most linear terms in y); the process conserves particle number; and the kernel is approximately separable in energy and angle. When all three hold, G_ξ is affine and the diagnostic vanishes.

Neutrino–nucleon elastic scattering ($\nu + N \rightarrow \nu + N$) is the paradigmatic Teff-compatible process: $\Delta E/E \sim E_\nu/M_N \approx 0.01\text{--}0.03$ for supernova energies, and the isoenergetic limit gives exact separability [46, 47]. Thomson scattering provides the photon-sector analog (Sec. V): zero energy transfer in the electron rest frame makes G affine at linear order, with the second-order correction scaling as the angular variance of Θ . Charged-current absorption with $\sigma \propto E^2$ also preserves the spectral shape when the cross section is smooth across the kinematic range [46, 48].

Neutrino–electron scattering ($\nu + e^- \rightarrow \nu + e^-$) is the clearest failure case: $E_\nu/m_e \sim 20\text{--}60$ for supernova neutrinos, the kernel is non-separable, and the $n = 2$ Laguerre mode dominates the diagnostic signal [49]. Pair processes ($e^+e^- \rightarrow \nu\bar{\nu}$) break the ansatz through a bilinear spectral coupling and number-changing production. In the photon sector, bremsstrahlung yields $G \propto y^{-1}$ —not affine—exciting all Laguerre modes. Compton scattering beyond the Thomson limit provides a quantitative boundary: the Teff representation holds to $\lesssim 1\%$ spectral accuracy for Comptonisation parameter $|y_C| \lesssim 0.005$ [50].

The diagnostic absorbs the cumulative effect of all active collision channels into a single scalar $\mathcal{D}_{\geq 2}$ (53), so the upgrade decision is made on the composite residual rather than on a per-process basis. We do not claim that the Teff representation is valid for all collision physics; we provide a computable criterion that monitors collision-side tangency process by process and timestep by timestep (Proposition 7 delimits the scope of this diagnostic).

B. Relation to existing methods

To place the framework among existing reduced kinetic descriptions, we summarise the structural comparison—to our knowledge—with the four closest methods. Levermore’s entropy closure hierarchy [12, 51] produces exponential-family distributions via global entropy optimisation, guaranteeing hyperbolicity and realisability at the cost of solving a constrained optimisation at each spacetime point; the Teff representation shares the exponential-family structure but replaces the global optimisation by an explicit per-direction Planckian closure, avoiding the boundary conditioning issues documented by Junk [15] and Hauck, Levermore, and Tits [16]. The partial-moment closures of Frank, Dubroca, and Klar [17]

share the per-direction philosophy but discretise the angular domain into patches and solve a per-patch entropy optimisation, whereas the Teff field $\Theta(e)$ is a continuous PSTF expansion with an explicit functional form. The Gorban–Karlin defect of invariance [18, 20] is the direct ancestor of the tangency diagnostic at the level of abstract principle; the present work instantiates it for per-direction exponential families in radiation transport with the operational distinctions noted in Sec. V B: a radiation-specific manifold, a model-switch response rather than Newton correction, and a computationally cheap scalar evaluation. The φ -divergence closures of Abdelmalik, Cai, and Pichard [21] represent the most recent application of divergence-based projections to the radiative transfer equation; their construction uses global moment constraints, whereas the present work uses per-direction families whose Pythagorean decomposition yields the tangency diagnostic as the Fisher-orthogonal residual. In each comparison, the Teff framework trades generality (it cannot represent arbitrary spectral shapes) for three operational gains: an explicit closure requiring no optimisation, a computable diagnostic identifying when collisions remain tangent to the representation manifold, and a Pythagorean decomposition connecting that diagnostic to the thermodynamic cost of the reduction.

Across the series, the load-bearing structure is not the specific one-field or two-field ansatz, but the positive Fisher-weight Hilbert geometry of a finite-dimensional exponential-family tangent space. The collision-side tangency criterion, the entropy curvature identity, and the Pythagorean residual decomposition are all instances of this common structure; what changes between papers is the dimension and parameterisation of the manifold.

C. Limitations

Five scope boundaries delimit the present work; each points to a concrete extension inheriting the diagnostic infrastructure. First, all benchmarks use controlled collision operators chosen for analytical tractability rather than microphysical realism; the formal theorem scope exceeds the benchmark coverage for Fermi–Dirac and Bose–Einstein statistics (Table II). Second, the benchmarks operate in axisymmetric angular configurations, so the full tensor coupling patterns of the 3D PSTF expansion are not exercised. Third, the formal results assume $\mu \leq 0$ (non-degenerate regime for fermions); the degenerate extension $\mu > 0$ requires changing the diagnostic energy variable from the shifted coordinate $y = (E - \mu)/(k_B T_0)$ to the strictly positive $x := E/(k_B T_0) > 0$, moving μ back into the exponential-family intercept $\alpha = -\eta/\Theta$ (where $\eta := \mu/(k_B T_0)$). Since $y = x - \eta$, the tangent space $\text{span}\{1, x\} = \text{span}\{1, y\}$ and the affine tangency criterion survive unchanged; what must be rebuilt is the Laguerre basis and inner product, now on $x \in (0, \infty)$ rather than $y \in (0, \infty)$. This coordinate change also

makes the per-ray Teff ansatz a one-dimensional curve $\alpha = -\eta\beta$, $\beta = 1/\Theta(e)$ inside the two-parameter family, restoring the framework of Sec. VIA. The exact boost law (Theorem 3, $\mu = 0$) does not extend: for $\mu \neq 0$ a direction-dependent shift appears (Proposition 4) that requires a joint Teff–fugacity degree of freedom. Fourth, the projected Teff hierarchy has not been evolved as a spatial transport solver; the numerical questions this entails (hyperbolicity away from isotropy, realizability-preserving discretisation, streaming–collision stiffness) are deferred to future work. Fifth, the formalism treats one particle species at a time; cross-species coupling (e.g. $\nu_e - \bar{\nu}_e$ through charged-current and pair processes) is deferred to future work.

D. Outlook

The formal results of this paper assume $\mu \leq 0$ and treat the one-field Teff-only regime. Four extensions build directly on the present machinery.

Paper II adds a direction-dependent chemical potential $\eta(\hat{e})$ to the Teff manifold, motivated by particle-number conservation in degenerate systems. Working in the strictly positive coordinate $x = E/(k_B T_0) > 0$ (which removes the domain pathology of the shifted variable y at $\mu > 0$), Paper II establishes that a scalar chemical potential cannot reproduce the joint energy–number moment structure generated by a direction-dependent η : the chemical-potential dipole is a genuinely non-redundant degree of freedom.

Paper III develops the entropy geometry and tangency structure of the two-field (Θ, η) manifold. The Gram matrix of the tangent-space inner product—which certifies parameter identifiability in Paper II—is shown to be identically the negative of the entropy second variation restricted to the manifold tangent space. The tangency diagnostic of the present paper survives on the extended manifold, with a monotone reduction from the one-field case.

The degenerate extension ($\mu > 0$) is particularly straightforward once the diagnostic is recast in x -coordinates (see the third limitation above): the geometric machinery (entropy-normalised collision field, affine tangency criterion, Pythagorean entropy-gap decomposition) carries over unchanged, and only the basis domain and weight must be adapted. The genuinely delicate regime is the sharp $T \rightarrow 0$ Fermi limit, where $f(1-f)$ concentrates near the Fermi surface and numerical conditioning—not formal geometry—becomes the limiting factor.

The projected hierarchy charts a path toward standalone transport solvers; benchmarks against M_1 and Monte Carlo references with realistic neutrino opacities, spatial discretisation analysis, and multi-species coupling constitute the program of subsequent work. In photon radiative transfer the Teff representation can serve as an angular-compression layer within each energy group,

with the diagnostic operating per group; in the quark-gluon plasma the 1+3 covariant formulation maps directly onto the Israel–Stewart framework, though dedicated calculations are needed to assess whether the diagnostic retains practical utility in that setting.

Where conventional closures promise an answer everywhere and deliver varying quality without warning, the Teff framework promises an answer only within the collision-tangent regime (Proposition 7) and delivers a computable signal when collisions begin to generate spectral distortions outside the per-ray equilibrium family.

All formal results in Paper I are per-ray, instantaneous, and collision-side statements for the one-field manifold under $\mu \leq 0$. They do not by themselves provide global kinetic adequacy, finite-time error control, or a degenerate- $\mu > 0$ extension.

Appendix A: Proofs and projection identities

1. Proof of Theorem 6

The argument proceeds in four steps. Steps 1–3 establish the equivalence between tangency and affinity of G_ξ ; Step 4 translates affinity into the vanishing of higher Laguerre modes.

Step 1 (Tangent vectors). The per-ray family (47) with $\phi = \alpha + \beta y$ gives $\partial_\phi f_\xi = -f_\xi(1 + \xi f_\xi)$ via the identity $e^\phi = (1 + \xi f_\xi)/f_\xi$. Since $\partial_\alpha \phi = 1$ and $\partial_\beta \phi = y$,

$$\partial_\alpha f_\xi = -f_\xi(1 + \xi f_\xi), \quad \partial_\beta f_\xi = -y f_\xi(1 + \xi f_\xi). \quad (\text{A1})$$

Step 2 (Tangency $\Rightarrow G_\xi$ affine). Tangency means $C = A(e)\partial_\alpha f + B(e)\partial_\beta f$ for direction-dependent A, B . By (A1), $C = -(A + By) f_\xi(1 + \xi f_\xi)$. Since $f_\xi(1 + \xi f_\xi) > 0$ on $(0, \infty)$ (hypothesis H3), $G_\xi := C/[f_\xi(1 + \xi f_\xi)] = -(A + By)$ —affine in y .

Step 3 (G_ξ affine $\Rightarrow$ tangency). If $G_\xi = a(e) + b(e)y$, then $C = G_\xi \cdot f_\xi(1 + \xi f_\xi) = -(A + By) f_\xi(1 + \xi f_\xi)$ with $A = -a$, $B = -b$, which is a linear combination of the tangent vectors (A1).

Step 4 (Affinity $\Leftrightarrow c_{n \geq 2} = 0$). By hypothesis (H5), $G_\xi(\cdot, e) \in \mathcal{H}_*$ and admits the expansion $G_\xi = \sum_{n \geq 0} c_n L_n^{(-1)}(y)$. Since $L_0^{(-1)} = 1$ and $L_1^{(-1)} = -y$, G_ξ affine in $y \Leftrightarrow G_\xi \in \text{span}\{L_0, L_1\} = \text{span}\{1, y\} \Leftrightarrow c_n = 0$ for all $n \geq 2$. The orthogonality of $\{L_n^{(-1)}\}$ under $\langle \cdot, \cdot \rangle_*$ (43) ensures uniqueness of the decomposition. $\square$

BE regularity near $y = 0$. For $\xi = +1$ with $\mu \leq 0$, $f \sim \Theta/y \rightarrow \infty$ as $y \rightarrow 0^+$ and $f(1+f) \sim \Theta^2/y^2$. For UU operators, the gain–loss form gives $G_{+1}(y_1) = \int W[f_3 f_4(1+f_2) - f_2(1+f_3)(1+f_4)] + \mathcal{O}(f_1^{-1})$, whose leading term is f_1 -independent and hence bounded as $y_1 \rightarrow 0^+$ for any bounded scattering kernel W . For BGK isotropisation, $G \sim (1-\Theta)y/(\tau\Theta^2) \rightarrow 0$ linearly. No analogous boundary subtlety arises for MB or FD with $\mu \leq 0$.

2. Proof of Theorem 9

The proof applies the Pythagorean theorem for Bregman divergences on dually flat manifolds to the per-ray exponential family. We verify the three hypotheses of the standard result [43, 44].

(i) *Strict convexity.* $\eta''_\xi(f) = 1/[f(1+\xi f)] > 0$ on the physical domain (hypothesis H1 of Theorem 9); hence η_ξ is a strictly convex Bregman generator and $D_\xi(f\|g) := \eta_\xi(f) - \eta_\xi(g) - \eta'_\xi(g)(f-g) \geq 0$ with equality iff $f = g$.

(ii) *e-projection structure.* At each direction e , the reduction map $\mathcal{R}$ performs the Bregman e-projection of $f(\cdot, e)$ onto the two-parameter per-ray manifold $\mathcal{M}_{\text{ray}}$ (47), minimising $D_\xi(f\|g)$ over $g \in \mathcal{M}_{\text{ray}}$. The sufficient statistics of $\mathcal{M}_{\text{ray}}$ are $(1, y)$ (corresponding to parameters (α, β)), so by the standard exponential-family result [44, 45], the e-projection matches both the per-ray normalisation $\int f dy$ and the per-ray first moment $\int y f dy$. Consequently, the entropy-normalised residual $\eta''_\xi(f^*) \cdot (f - f^*)$ has zero overlap with modes $n = 0$ and $n = 1$, and its Laguerre content lies entirely in $n \geq 2$.

(iii) *Pythagorean identity.* By [44, Theorem 3.1], for any $q \in \mathcal{M}_{\text{ray}}$,

$$D_\xi(f\|q) = D_\xi(f\|f^*) + D_\xi(f^*\|q), \quad (\text{A2})$$

where $f^* = f^*_{\text{ray}}$ is the per-ray e-projection. Setting $q = f^*$ and integrating over directions and energy with the on-shell measure gives $S_{\text{red}} - S_{\text{kin}} = k_B \int d\Pi D_\xi(f\|f^*) \geq 0$, with equality iff $f = f^*$ a.e. $\square$

3. Realizability proofs

Proof of Proposition 14. $\Theta(e) = 1 + T_{ab}e^ae^b$ with e a unit vector. By the Rayleigh quotient, $\min_e T_{ab}e^ae^b = \lambda_{\min}(T)$. Hence $\Theta > 0$ for all e iff $1 + \lambda_{\min} > 0$. $\square$

Proof of Proposition 15. $\Theta(\mu) = 1 + A\mu + Q(\mu^2 - \frac{1}{3})$ is a quadratic in $\mu \in [-1, 1]$ with leading coefficient Q .

Case $Q > 0$. Interior critical point $\mu^* = -A/(2Q)$ lies in $[-1, 1]$ iff $|A| \leq 2Q$. When interior, $\Theta(\mu^*) = 1 - Q/3 - A^2/(4Q)$; positivity requires $3A^2 + 4Q^2 < 12Q$. When $|A| > 2Q$, the minimum is at $\mu = -\text{sgn}(A)$: $\Theta_{\min} = 1 - |A| + 2Q/3 > 0$ iff $|A| < 1 + 2Q/3$.

Case $Q \leq 0$. The interior point is a maximum; minima at $\mu = \pm 1$ give $\Theta(\pm 1) = 1 \pm A + 2Q/3$, both positive iff $|A| < 1 + 2Q/3$.

C^1 junction. At $|A| = 2Q$ ($Q > 0$), implicit differentiation of the ellipse $6A dA + (8Q - 12) dQ = 0$ at $(|A|, Q) = (3/2, 3/4)$ gives $dA/dQ = 2/3$, matching the line slope. $\square$

Proof of Lemma B.2. By Cauchy-Schwarz, $|T_a e^a| \leq |T_a|$. By the Rayleigh quotient, $|T_{ab}e^ae^b| \leq \|T_{ab}\|_{\text{op}}$. The triangle inequality gives $\Theta(e) \geq 1 - |T_a| - \|T_{ab}\|_{\text{op}} > 0$. Sharpness: at $T_a = r\hat{n}_a$, $T_{ab} = -s\hat{n}_{(a}\hat{n}_{b)}$ with $r + 2s/3 = 1$, $\Theta(-\hat{n}) = 0$. $\square$

Appendix B: Projection identities

1. Laguerre energy-mode projection

The generalised Laguerre polynomials $L_n^{(\alpha)}$ with $\alpha = -1$ satisfy orthogonality on $(0, \infty)$ with weight $w(y) = y^{-1}e^{-y}$:

$$\begin{aligned} \langle L_m^{(-1)}, L_n^{(-1)} \rangle_* &:= \int_0^\infty \frac{dy}{y} e^{-y} L_m^{(-1)}(y) L_n^{(-1)}(y) \\ &= \frac{\delta_{mn}}{n} \quad (n \geq 1). \end{aligned} \quad (\text{B1})$$

The inner product (44) defines the Hilbert space $\mathcal{H}_*$ used in Theorem 6, hypothesis (H5). The $n = 0$ mode $L_0^{(-1)} = 1$ is not square-integrable under this weight ($\|L_0\|_*^2 = \int_0^\infty y^{-1}e^{-y} dy = +\infty$), so c_0 cannot be extracted by the orthogonality formula. For $n \geq 1$, the Laguerre coefficients of the entropy-normalised collision field are

$$c_n^\xi(e) = n \int_0^\infty \frac{dy}{y} e^{-y} G_\xi(y, e) L_n^{(-1)}(y). \quad (\text{B2})$$

The lowest explicit polynomials are $L_0^{(-1)} = 1$, $L_1^{(-1)} = -y$, $L_2^{(-1)} = \frac{1}{2}y^2 - y$.

Lemma B.1 (L_0 leakage and regression equivalence). *Direct Laguerre projection of G_ξ onto modes $n \geq 2$ using (46) produces structurally contaminated coefficients whenever $c_0 \neq 0$, because $L_0^{(-1)} = 1$ is not orthogonal to $L_n^{(-1)}$ for $n \geq 2$ under $\langle \cdot, \cdot \rangle_*$:*

$$\langle L_0, L_n \rangle_* = \int_0^\infty \frac{dy}{y} e^{-y} L_n^{(-1)}(y) = -\frac{1}{n}, \quad n \geq 1. \quad (\text{B3})$$

Consequently, formula (46) applied to an exactly affine $G = a + by$ yields $c_2^{\text{extracted}} = -a$, not 0. This is an exact algebraic contamination, not a truncation artifact. The regression approach—fitting $G_\xi = a + by$ by weighted least squares in $\langle \cdot, \cdot \rangle_*$ and computing $d_{\geq 2} = \|G_\xi - a - by\|_*$ —eliminates this leakage by construction and achieves machine-precision tangency for MB statistics.

Proof. Write $G = c_0 L_0 + \sum_{n \geq 1} c_n L_n^{(-1)}$ with true coefficients c_n . Applying (46): $c_n^{\text{extr}} = n \langle L_n, G \rangle_* = n[c_0 \langle L_n, L_0 \rangle_* + c_n/n] = c_n - c_0$. For $G = a + by$ (so $c_0 = a$, $c_1 = -b$, $c_{n \geq 2} = 0$): $c_2^{\text{extr}} = 0 - a = -a$. The regression approach computes $\min_{a,b} \|G - a - by\|_*$, which equals $\|\sum_{n \geq 2} c_n L_n\|_*$ by orthogonality of $\{L_n\}_{n \geq 1}$; this uses the true c_n and is uncontaminated. $\square$

2. Sufficient realizability bound (mixed angular orders)

Lemma B.2 (Sufficient realizability bound). *Let T_a be a spatial vector and $T_{ab} = T_{\langle ab \rangle}$ a PSTF tensor. Define*

$\Theta(e) = 1 + T_a e^a + T_{ab} e^a e^b$. Then

$$|T_a| + \|T_{ab}\|_{\text{op}} < 1 \implies \Theta(e) > 0 \text{ for all unit } e, \quad (\text{B4})$$

where $|T_a| \equiv (T_a T^a)^{1/2}$ and $\|T_{ab}\|_{\text{op}} = \max(|\lambda_{\text{max}}(T)|, |\lambda_{\text{min}}(T)|)$. This bound is sharp: for any $|T_a| + \|T_{ab}\|_{\text{op}} = 1$ there exists a configuration with $\min_e \Theta(e) = 0$.

The proof is in Appendix A 3. Discrete-data computation steps and positivity-limiter analysis for the one-mode upgrade are deferred to future work.

3. Perturbative boost algebra (Proof of Proposition 5)

We derive Eqs. (29)–(32) from the exact law (27). Throughout, v^a is the boost velocity with $v^a u_a = 0$ and $|v| \ll 1$.

Lemma B.3 (PSTF angular averages). *For a unit spatial vector e^a on S^2 ,*

$$\langle e^a e^b \rangle_\Omega = \frac{1}{3} h^{ab}, \quad (\text{B5})$$

$$\langle e^a e^b e^c e^d \rangle_\Omega = \frac{1}{15} (h^{ab} h^{cd} + h^{ac} h^{bd} + h^{ad} h^{bc}). \quad (\text{B6})$$

All odd-rank averages vanish. For a PSTF tensor $T_{ab} = T_{\langle ab \rangle}$ and any spatial vector w^c ,

$$\langle T_{ab} e^a e^b \rangle_\Omega = 0, \quad T_{ab} w_c \langle e^a e^b e^c e^d \rangle_\Omega = \frac{2}{15} T^d{}_c w^c, \quad (\text{B7})$$

where the first identity follows from trace-freeness and $\langle e^a e^b \rangle_\Omega = \frac{1}{3} h^{ab}$, and the second from (B6) with $T^a{}_a = 0$ eliminating one of the three pairings.

Proof of Proposition 5. Start from the exact transformation law (27):

$$\tilde{\Theta}(\tilde{e}) = \frac{T_0}{\tilde{T}_0} \Theta(e(\tilde{e})) D(e(\tilde{e})), \quad \frac{\tilde{T}_0}{T_0} = \langle \Theta D \rangle_\Omega. \quad (\text{B8})$$

Step 1: Expand D and the aberration map. The Doppler factor for massless particles is $D(e) = \gamma(1 + v_a e^a)$. To the required order,

$$D(e) = 1 + v_a e^a + \frac{1}{2} (v_a e^a)^2 - \frac{1}{2} v^2 + \mathcal{O}(v^3). \quad (\text{B9})$$

The inverse aberration map satisfies $e^a(\tilde{e}) = \tilde{e}^a - v^a + (v_b \tilde{e}^b) \tilde{e}^a + \mathcal{O}(v^2)$; substituting into $\Theta(e(\tilde{e}))$ and using the PSTF expansion $\Theta(e) = 1 + T_a e^a + T_{ab} e^a e^b + \dots$ gives, to $\mathcal{O}(v)$:

$$\Theta(e(\tilde{e})) = 1 + T_a \tilde{e}^a + T_{ab} \tilde{e}^a \tilde{e}^b - T_a v^a + (T_a v_b + T_{ab} v^b) \tilde{e}^a \cdot \delta + \mathcal{O}(v^2), \quad (\text{B10})$$

where the δ -terms encode the aberration correction at each rank.

Step 2: Form the product $\Theta \cdot D$ and average. The numerator of $\tilde{\Theta}$ is $N(\tilde{e}) := \Theta(e) D(e)$ evaluated along

the aberration-mapped direction. Collecting terms by angular rank in $\tilde{e}$ and using the averages of Lemma B.3:

$$\begin{aligned} \langle N \rangle_\Omega &= 1 + \frac{1}{3} T_a v^a + \frac{1}{6} v^2 - \frac{1}{2} v^2 + \mathcal{O}(v^3) \\ &= 1 + \frac{1}{3} T_a v^a - \frac{1}{3} v^2 + \mathcal{O}(v^3), \end{aligned} \quad (\text{B11})$$

where the $+\frac{1}{6}v^2$ arises from $\langle (v_a e^a)^2 \rangle_\Omega = \frac{1}{3} v^2$ multiplied by $\frac{1}{2}$, and $-\frac{1}{2}v^2$ is the constant part of the Doppler expansion. This is the denominator $\tilde{T}_0/T_0 = \langle N \rangle_\Omega$, yielding Eq. (29) after noting $-\frac{1}{3}v^2 = -\frac{1}{6}v_a v^a \cdot 2$ and collecting the sign.

Step 3: Extract the dipole. The rank-1 PSTF projection of $\tilde{\Theta}(\tilde{e})$ is $\tilde{T}_d = 3 \langle N \tilde{e}_d \rangle_\Omega$ at leading order in $(T_0/\tilde{T}_0) \approx 1$. Three $\mathcal{O}(v)$ terms in $N(\tilde{e})$ contribute a rank-1 projection after angular averaging:

- (a) Doppler factor alone: $v_c \langle \tilde{e}^c \tilde{e}_d \rangle_\Omega = \frac{1}{3} v_d$.
- (b) Aberration of T_{ab} : the inverse aberration shifts $T_{ab} e^a e^b \rightarrow T_{ab} \tilde{e}^a \tilde{e}^b (1 + 2v_c \tilde{e}^c) - 2T_{ab} v^a \tilde{e}^b + \mathcal{O}(v^2)$. The factor $(1 + 2v_c \tilde{e}^c)$ generates a rank-3 term $2T_{ab} v_c \tilde{e}^a \tilde{e}^b \tilde{e}^c$, whose rank-1 projection via (B6) is $\frac{4}{15} T_{da} v^a$. The direct term $-2T_{ab} v^a \tilde{e}^b$ projects to $-\frac{2}{3} T_{da} v^a$.
- (c) $\Theta_2 \times D_1$ cross: $T_{ab} \tilde{e}^a \tilde{e}^b \cdot v_c \tilde{e}^c$ projects via (B7) to $\frac{2}{15} T_{da} v^a$.

Summing (b) and (c): $(\frac{4}{15} - \frac{10}{15} + \frac{2}{15}) T_{da} v^a = -\frac{4}{15} T_{da} v^a$. Multiplying by 3:

$$\tilde{T}_d = T_d + v_d - \frac{4}{5} T_{da} v^a + \mathcal{O}(v^2), \quad (\text{B12})$$

which is Eq. (30). The coefficient $-4/5 = 3 \times (-4/15)$ is the net result of the three competing contributions.

Step 4: Extract the quadrupole. The rank-2 PSTF projection uses the identity $\tilde{T}_{ab} = \frac{5}{2} \cdot 3 \langle N \tilde{e}_a \tilde{e}_b \rangle_\Omega$ at leading order. The product $N \tilde{e}_a \tilde{e}_b$ has two sources of rank-2 content at $\mathcal{O}(v^0)$ and $\mathcal{O}(v^1)$: the original T_{ab} (from Θ), and the cross terms $T_a v_b$ and $v_a v_b$ (from $\Theta \cdot D$ and D^2). After PSTF projection:

$$\tilde{T}_{ab} = T_{ab} + 2v_{\langle a} T_{b \rangle} + v_{\langle a} v_{b \rangle} + \mathcal{O}(v^3), \quad (\text{B13})$$

which is Eq. (31). The coefficient 2 in $2v_{\langle a} T_{b \rangle}$ is the standard STF product of a rank-1 and a rank-1 tensor promoted to rank 2.

Step 5: Extract the octupole. The rank-3 projection picks up the product of the existing T_{ab} with the boost velocity v_c :

$$\tilde{T}_{abc} = 3T_{\langle ab} v_{c \rangle} + \mathcal{O}(v^2). \quad (\text{B14})$$

This is Eq. (32); the factor 3 is the number of distinct STF pairings of one vector against a rank-2 PSTF tensor. $\square$

Numerical check. Direct evaluation of the exact law (27) on a 1024-point Gauss–Legendre grid, followed by PSTF projection, recovers the above coefficients to $\mathcal{O}(10^{-5})$ relative accuracy for $\beta \leq 0.01$ (Sec. III D).

4. Conditional coercivity of the angular mass matrix

Proposition 16 (Conditional coercivity). *Fix a finite angular basis $\{\Psi_I(e)\}_{I=1}^N$ on S^2 , and suppose that for the chosen weighted hierarchy the symmetrised mass matrix can be written as*

$$M_{IJ}^{(s)} = \int_{S^2} d\Omega Q(x, e) \Psi_I(e) \Psi_J(e), \quad (\text{B15})$$

with $Q(x, e) = q(x) \Theta(x, e)^p$, $q(x) > 0$, $p \in \mathbb{R}$, and $\Theta(x, e) \geq \Theta_{\min}(x) > 0$ for all e . Then $M^{(s)}$ is symmetric positive definite. More precisely, writing $\Theta_*^p := \inf_{e \in S^2} \Theta(x, e)^p$ (which equals $\Theta_{\min}^p$ for $p \geq 0$ and $\Theta_{\max}^p$ for $p < 0$, by monotonicity of $t \mapsto t^p$),

$$V^I M_{IJ}^{(s)} V^J \geq q(x) \Theta_*^p \int_{S^2} d\Omega \left(\sum_I V^I \Psi_I \right)^2, \quad (\text{B16})$$

and the right-hand side is strictly positive for every nonzero V if the $\{\Psi_I\}$ are linearly independent in $L^2(S^2, d\Omega)$. Hence $M^{(s)}$, and therefore M , is invertible.

Proof. Symmetry is immediate from the definition. For any nonzero coefficient vector V , define $\Psi_V(e) := \sum_I V^I \Psi_I(e)$. Then

$$V^I M_{IJ}^{(s)} V^J = \int_{S^2} d\Omega q \Theta^p \Psi_V(e)^2. \quad (\text{B17})$$

Since $q > 0$ and $\Theta^p \geq \Theta_*^p := \inf_e \Theta^p > 0$,

$$V^I M_{IJ}^{(s)} V^J \geq q \Theta_*^p \int_{S^2} d\Omega \Psi_V(e)^2. \quad (\text{B18})$$

If the angular basis is linearly independent, $\Psi_V \neq 0$ whenever $V \neq 0$, so the final integral is strictly positive. Thus $M^{(s)}$ is positive definite. $\square$

5. Counterexamples: scope boundaries of the diagnostic

Example 1 (Per-ray tangent but not Teff-manifold tangent). Consider a collision operator that, at every direction e , produces an entropy-normalised field with $c_{n \geq 2}(e) = 0$ (affine in y), so that $\mathcal{D}_{\geq 2} = 0$ and the per-ray manifold $\mathcal{M}_{\text{ray}}$ is tangent at every ray. However, suppose the $n = 0$ coefficient is direction-dependent: $c_0(e) \neq \text{const.}$ Then the collision drives a direction-dependent amplitude update that cannot be absorbed into a shared (T_0, μ) gauge. The collision is tangent to the product manifold $\bar{\mathcal{M}} = \prod_e \mathcal{M}_{\text{ray}}(e)$ but *not* to the global Teff manifold $\mathcal{M}_{\text{Teff}}$ (which imposes $\langle \Theta \rangle_\Omega = 1$). This illustrates that per-ray tangency ($c_{n \geq 2} = 0$) is necessary but not sufficient for full Teff-manifold tangency; the angular-consistency obstruction on $c_0(e)$ is a separate condition (Proposition 7(e), Remark V B).

Example 2 (Streaming-only departure). Start from a Teff state f_Θ with $\Theta(e) > 0$ and evaluate the collision field: suppose $\mathcal{D}_{\geq 2} = 0$ (collision is tangent). Now evolve one streaming step: $f(t+\Delta t) = f_\Theta - \Delta t e^a \partial_a f_\Theta + \mathcal{O}(\Delta t^2)$. The spatial gradient $\partial_a f_\Theta$ generally introduces angular content at ranks $\ell > L$ through the nonlinear composition $\Theta \mapsto \Phi(y/\Theta)$, so the evolved state may lie outside the L -truncated Teff manifold, yet $\mathcal{D}_{\geq 2} = 0$ provides no warning. This is the streaming blind spot: the collision-side diagnostic does not monitor angular-truncation failures generated by spatial gradients. A separate streaming-side proxy would be needed.

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